

CS-5114 (STRATEGIES FOR EMERGING TECHNOLOGY II)

- **Virtualization Technology**
- **Network and Cloud Security**
- **Internet of Things (IoT) and**
- **Blockchain Technology**

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BLOCKCHAIN TECHNOLOGY

- Introduction to Blockchain Technology
- Technical Concepts of Blockchain Technology
- Blockchain Architecture and Use Cases

What is Blockchain?

Definitions of Blockchain

- **Technically**, the blockchain is a **back-end database** that **maintains a distributed ledger** that can be inspected openly.
- **Business-wise**, the blockchain is **an exchange network for moving transactions**, value, assets between peers, without the assistance of intermediaries.
- **Legally**, speaking, the blockchain **validates transactions**, replacing previously trusted entities.

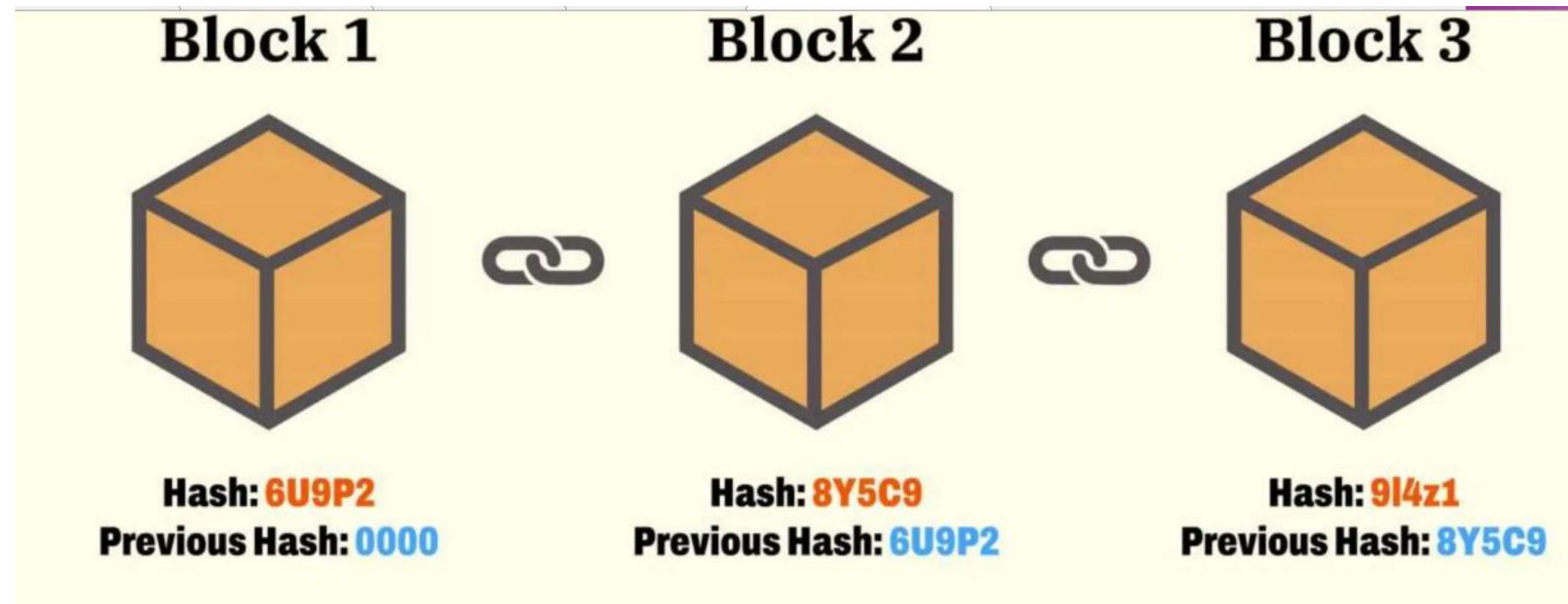
Blockchain capabilities = technical + business + legal

What is Blockchain?

- **Blockchain Technology** is a decentralized and distributed ledger system that allows multiple parties to record and store information in a secure, transparent, and tamper-resistant manner. The fundamental concept behind blockchain is to create a chain of blocks, where each block contains a list of transactions or records, and each block is linked to the previous one through a cryptographic hash.

What is Blockchain?

- **In simple words**, a blockchain is a chain of blocks that store information. Each block in the chain contains a **set of transactions**, and once the block is filled, it is added to the chain, forming a **permanent record of the transactions**. This chain is then **duplicated and distributed** across a network of computers, making it nearly **impossible to alter or delete information**.



What is Blockchain?

- Blockchain technology is a software; a protocol for the secure transfer of unique instances of value (e.g. money, property, contracts, and identity credentials) via the internet without requiring a third-party intermediary such as a bank or government.
 - Email over IP, Voice over IP, Money over IP



What is Blockchain?

- Blockchain $\neq$ Bitcoin
- Blockchain is not a cryptocurrency.
- Blockchain is the technology behind cryptocurrency, and this is just one use-case of the technology.
- Blockchain products for almost every industry, just like any other software. But only smarter and more reliable.
- Blockchain is a **decentralized ledger**, tracking digital assets on a **Peer-to-Peer network**.
- Each node (computer/server) connects to one another on the network, and each node **holds the same copy of the ledger**.

What is Blockchain?

- A ledger is a record book, which holds **details of every transaction** that takes place on the network. Each machine contains a copy of this ledger, in such a way that it **verifies each transaction**.
- **Example**, think of it as a bank statement. You and four other friends hold a copy of this same statement. When any of your friends make a transaction, the system will check the last statement and verify if it is the same in all the other four. Only once this background check is completed, can a new transaction take place. After the transaction is completed, it will be updated in all five ledgers. If anybody tries to manipulate their statement, it won't match the rest of the ledgers. This is sure-shot way of avoiding fraud. Now imagine this happening on thousands of machines, independently for every transaction. An attacker cannot manipulate so many devices simultaneously.

Reference link- <https://www.youtube.com/watch?v=yubzJw0uiE4>

How do block and chain, and how they are related?

- Think of the **book as Blockchain**, each page contains some data.
- Every individual page as a single block, and the data on this page as a Blockchain transaction.
- All these pages are in particular order and references to the previous page.
- It is essential not to delete any pages because then it can change the meaning of the entire agreement.
- If in case any page has been removed or deleted, it is easily identifiable because every page is numbered.
- Removing one page will corrupt the entire ledger.
- For example, if you remove page 65 from a 100-page book, then it will get noticed when somebody tallies it.
- Three questions arise from here.

How do block and chain, and how they are related?

- What does this have to do with digital assets or currencies?
- What if the page is not deleted and instead, the data on it has been tampered?
- Now, will we tally every block on each node for making a new transaction?

Reference Link-

Blockchain for Beginners by Blockchain Council, 2020

Understanding the Blockchain -<https://www.youtube.com/watch?v=r43LhSUUGTQ>

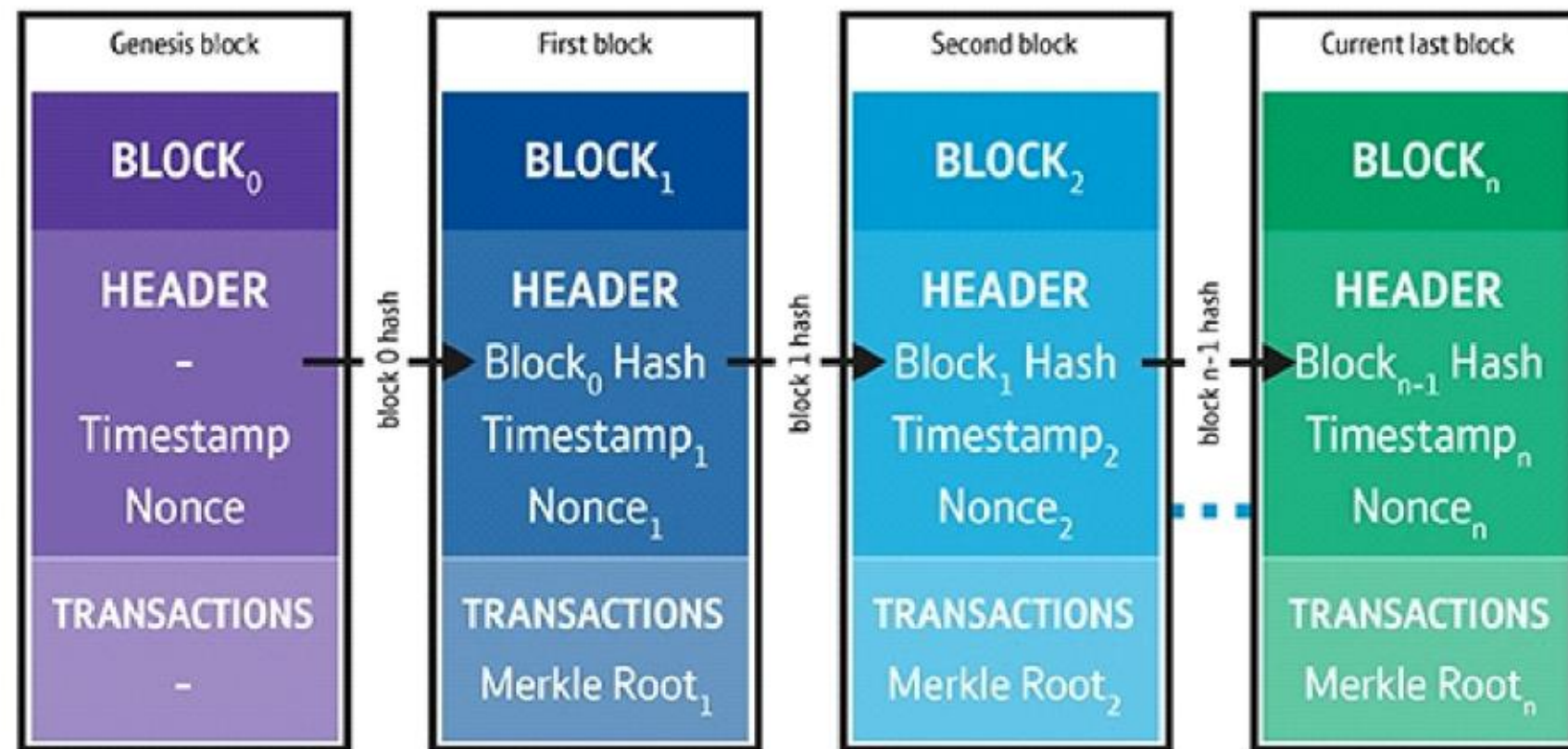
<https://www.youtube.com/watch?v=t5JGQXCTe3c>

<https://www.youtube.com/watch?v=19jOJk30eQs>

<https://www.youtube.com/watch?v=yubzJw0uiE4>

How blocks are linked together to form a blockchain

- The process starts with the creation of a block, which contains **transaction data, a timestamp, and a unique identifier called a hash**. This hash is crucial in connecting blocks together.
- When a new block is created, it includes the hash of the most recent block added to the chain.
- This forms a **link between the new block and the previous one**, creating a chain of blocks, hence the term "blockchain."

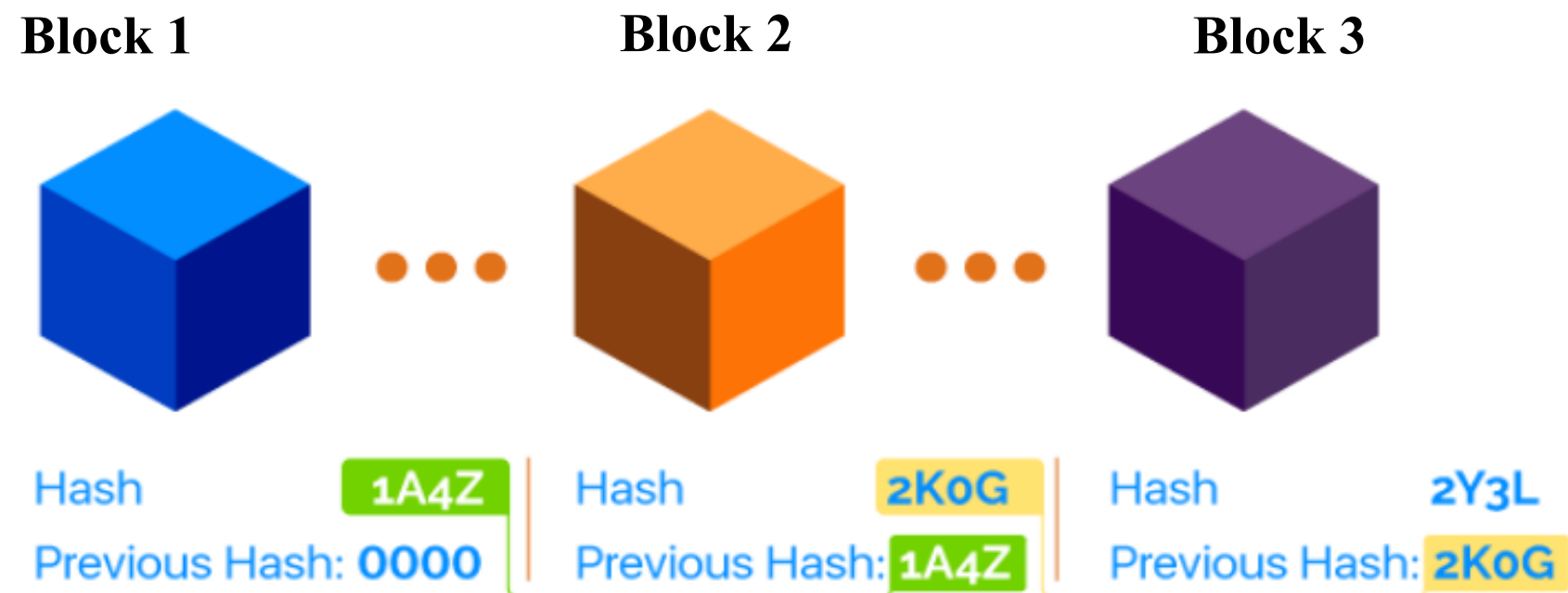


How blocks are linked together to form a blockchain

- Each block's hash is dependent on the information contained in the block itself, as well as the hash of the previous block.
- This implies that if anyone attempts to **alter the information in a block**, the **block's hash would change**, breaking the link with the following block.
- To restore this link, the intruder would need to alter the information in every subsequent block, which is virtually impossible given the computational power required.
- This unique linkage mechanism of blocks in a **blockchain provides a high level of security and immutability**.

How blocks are linked together to form a blockchain

- Block 1 is a record of only one transaction as distributed record.
- When block 2 want to be created , block 1 must be created the signature (cryptographic hash function)
- Block 1 output hash is same as the input hash of block2.
- Also, each block contains a hash of the previous block. For instance, if there are three blocks in a blockchain, block 3 will contain the hash of block2, and block 2 will contain the hash of block 1.



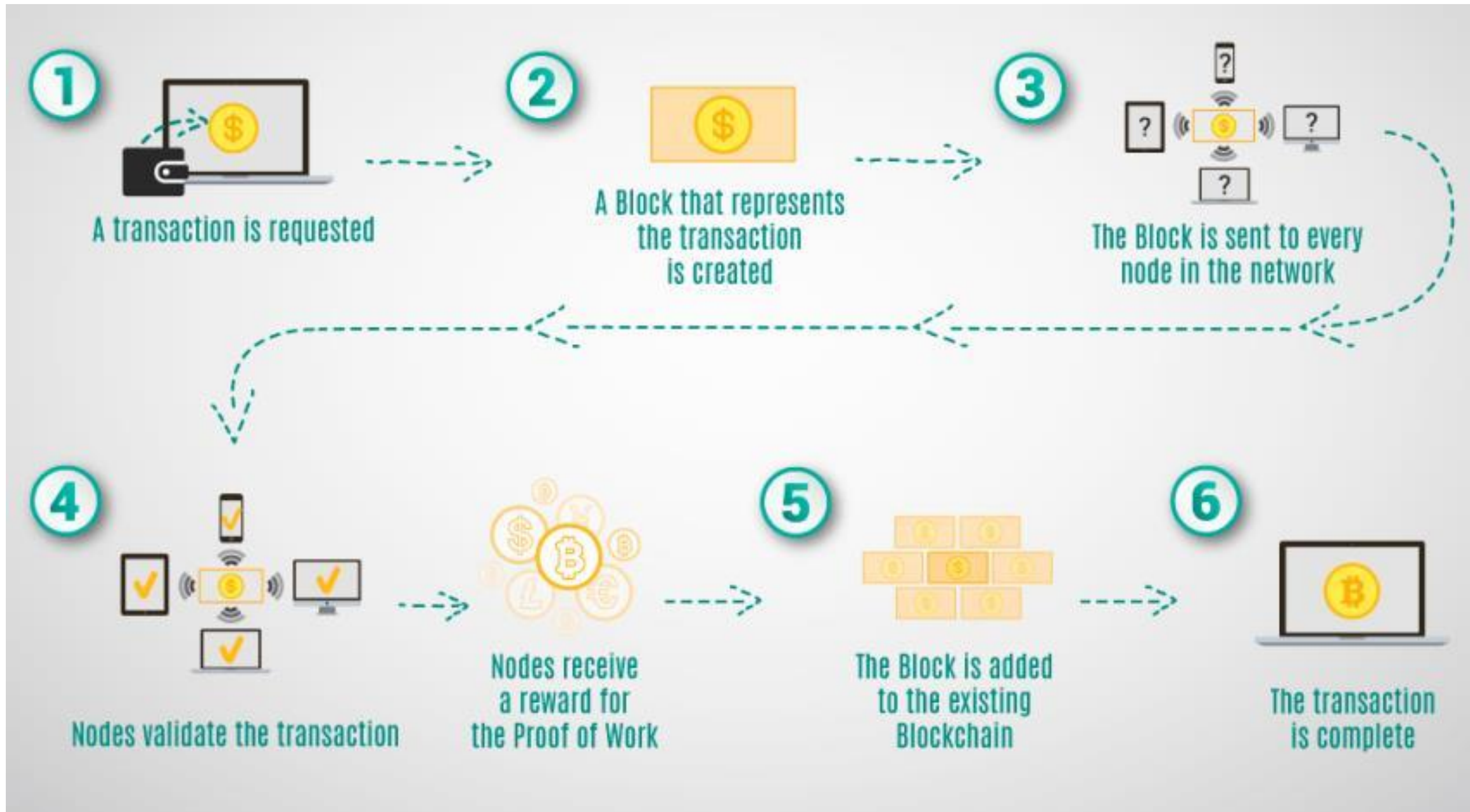
How blocks are linked together to form a blockchain

- If anyone changes the data in a single block, the hash of that particular block changes, but it also makes the whole chain invalid.



- A hash is a great tool for identifying attempts to change data in blocks. However, a hash algorithm alone is not enough to ensure the security of a blockchain. To mitigate attempts to corrupt the blockchain and to ensure security, blockchain technology also uses a process called [proof-of-work](#).

How blocks are linked together to form a blockchain



How a Blockchain works?

Blockchain is a combination of technologies:

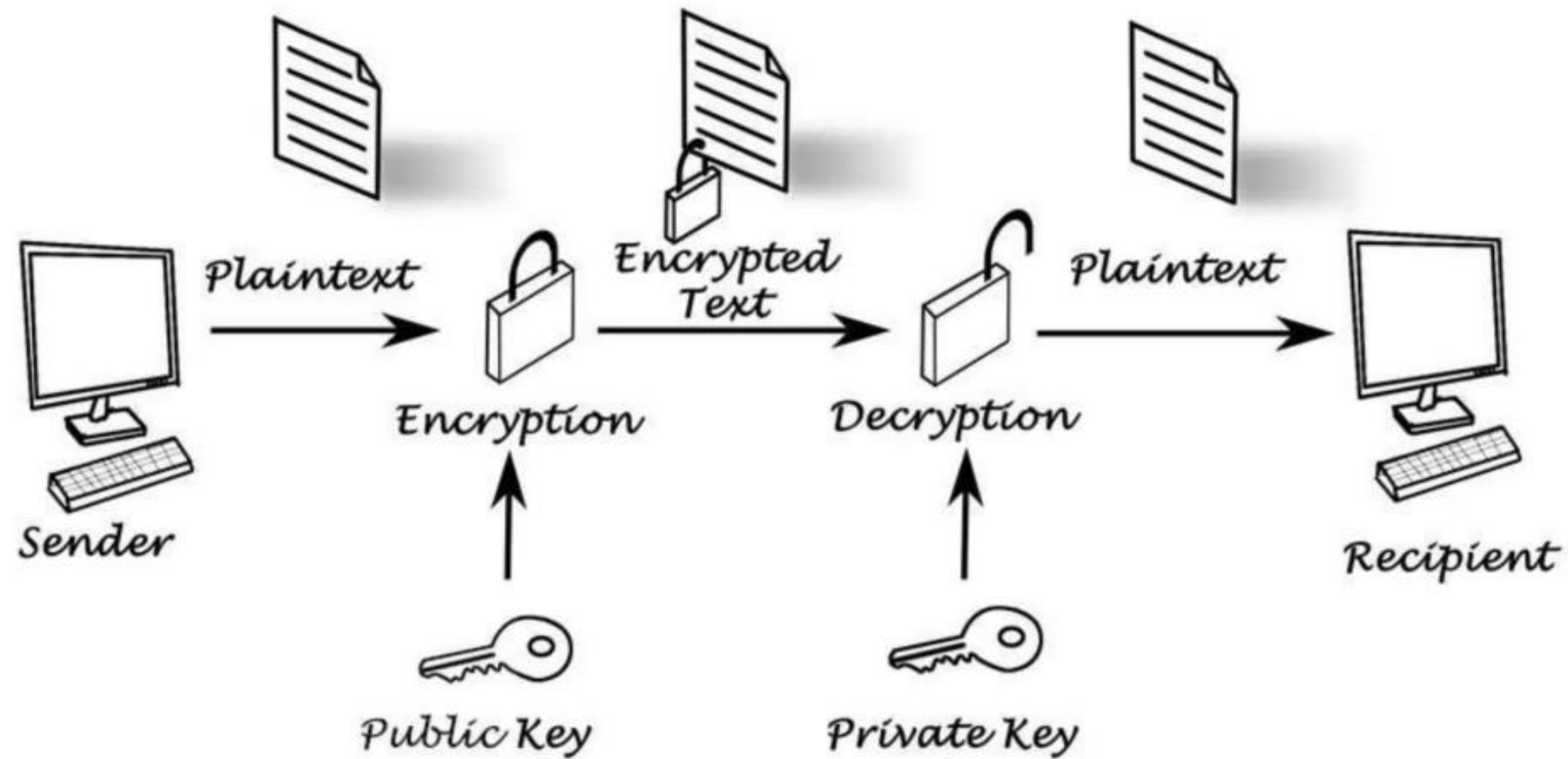
- Cryptographic keys
- A peer-to-peer network containing a shared ledger

A means of computing, to store the transactions and records of the network

What is cryptography ?

- In simple terms, it involves making information unreadable to anyone who doesn't have the proper key to access it
- **Cryptography keys consist of two keys** – Private key and Public key. These keys help in performing successful transactions between two parties. Each individual has these two keys, which they use to produce a secure digital identity reference. This secured identity is the most important aspect of Blockchain technology

Cryptographic Keys



Cryptographic Algorithm

A cryptographic algorithm is the **mathematical equation** used to scramble the plain text and make it unreadable. They are used for **data encryption, authentication and digital signatures**.

There are **three types** of cryptography:

- 1) Symmetric-key cryptography
- 2) Public-key or asymmetric cryptography
- 3) Hash-function

Cryptographic Algorithm

- 1) **Symmetric-key cryptography** - Both sender and receiver share a single key and the sender uses this key to encrypt plaintext. The cipher text is sent to the receiver, and the receiver can apply this same key to decrypt the message and recover the plain text from the sender.
- 2) **Public-key or asymmetric cryptography** –In public key cryptography (PKI), also known as asymmetric cryptography, there are two related keys called the public and private key. While the public key may be freely distributed, the paired private key must remain confidential. The public key is used for encryption and the private key is used for decryption.

Cryptographic Algorithm

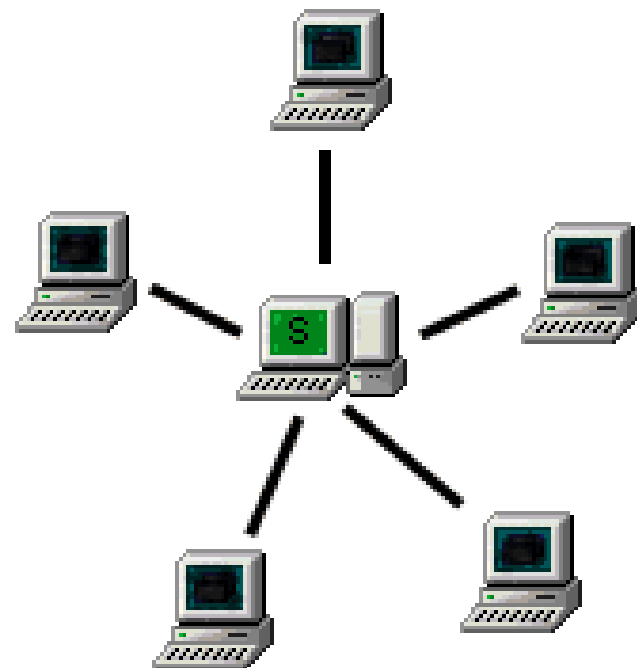
The most used asymmetric cryptography algorithms are RSA and ECC. TLS/SSL certificates frequently use RSA keys, and the recommended size of these keys is continually increasing (e.g. 1024 bit to 2048 bit) to maintain sufficient cryptographic strength. An alternative to RSA is ECC, which can offer the same level of cryptographic strength at much smaller key sizes, offering improved security with reduced computational and storage requirements.

3) Hash-function – No key is used in this algorithm, but a fixed-length value is calculated from the plaintext, which makes it impossible for the contents of the plaintext to be recovered. Hash functions are often used by computer systems to encrypt passwords. A popular hash algorithm is 256-SHA.

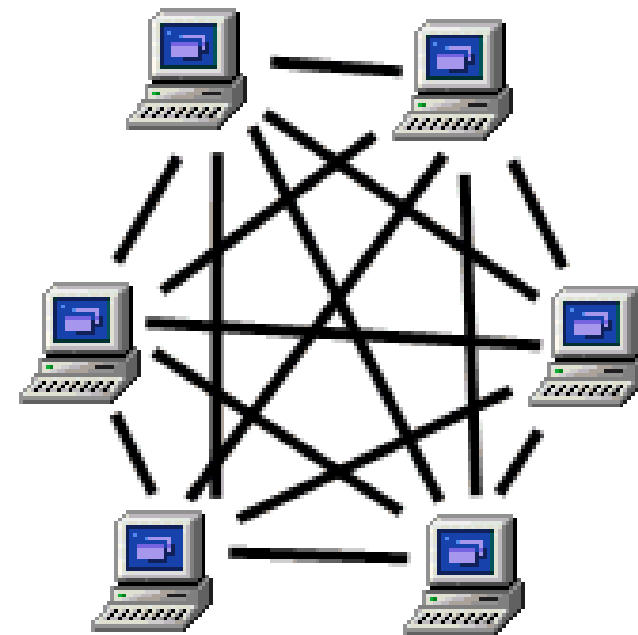
Peer-to-Peer Network

The term **Peer-To-peer (P2P)** refers to decentralized networks of interconnected computer systems **containing peers, or nodes**. All nodes are equal, and the exchange of data occurs without a central server — that is, each computer or node can act as both a file server and a client.

Server Based Network



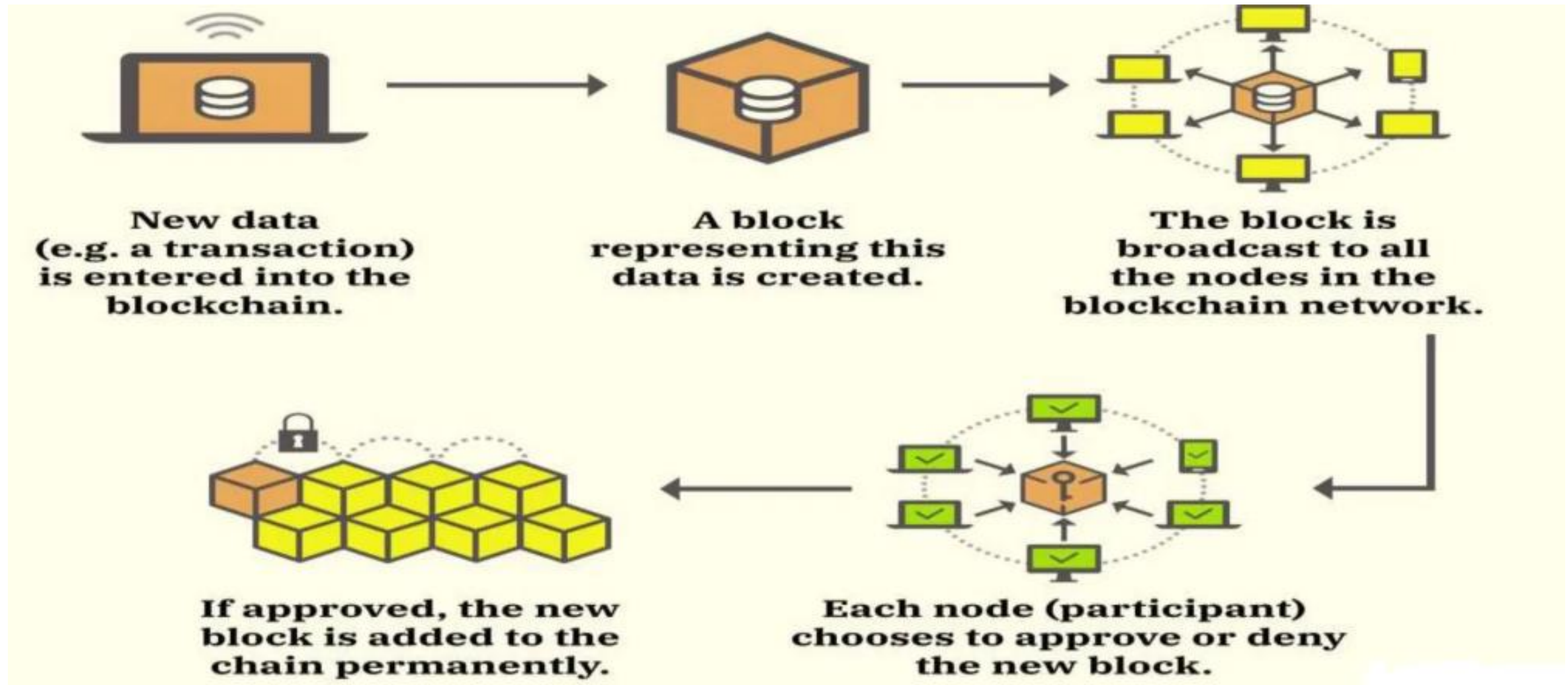
Peer to Peer Network



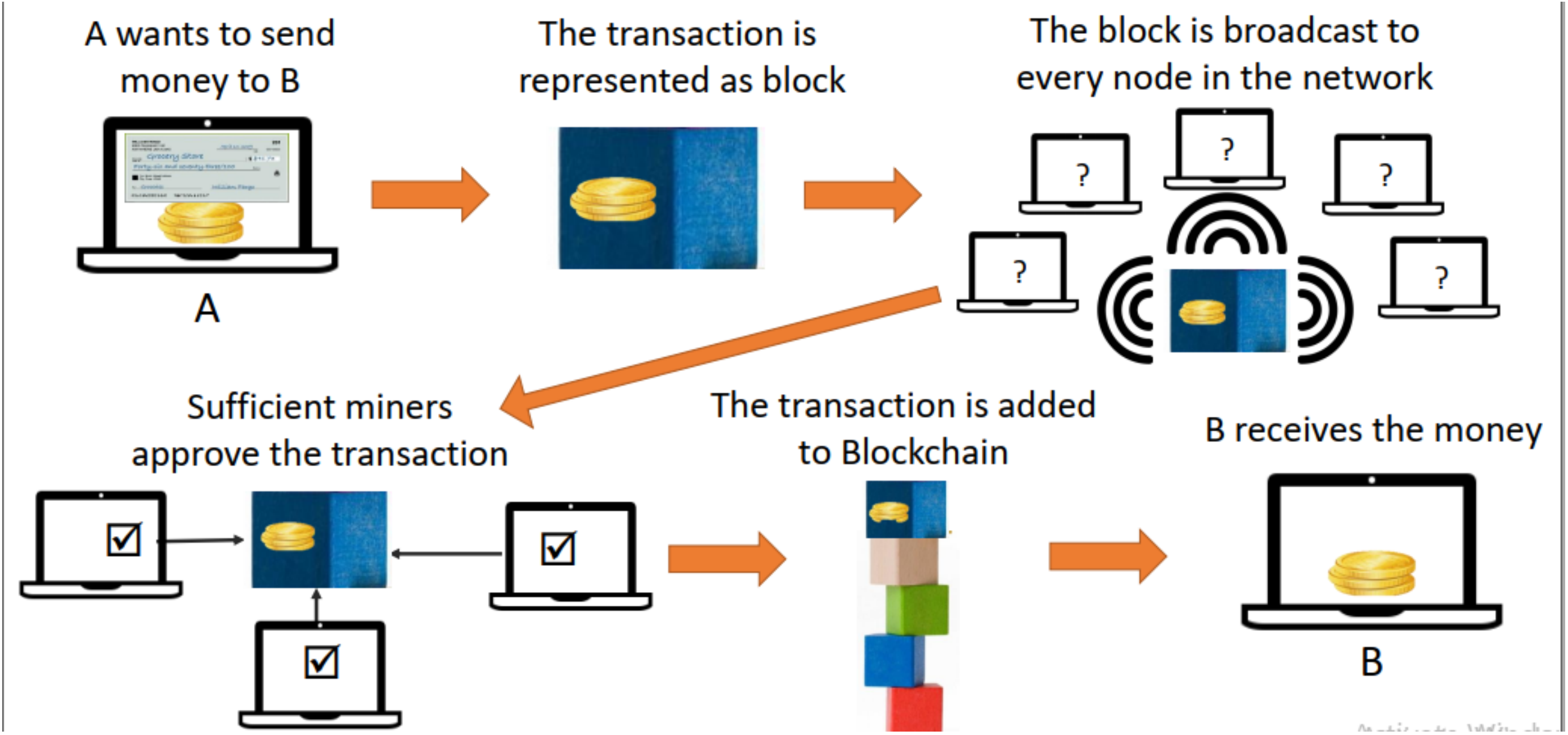
Peer-to-Peer Network

- A peer-to-peer (P2P) network is a fundamental part of blockchain technology that allows for the **secure and decentralized recording** of transactions.
- Blockchain is a P2P network that acts as a **decentralized ledger for one or more digital assets**, which refers to a decentralized peer-to-peer system where each computer keeps a complete copy of the ledger and verifies its authenticity with other nodes to guarantee the data is accurate.

Blockchain Process



Blockchain Process



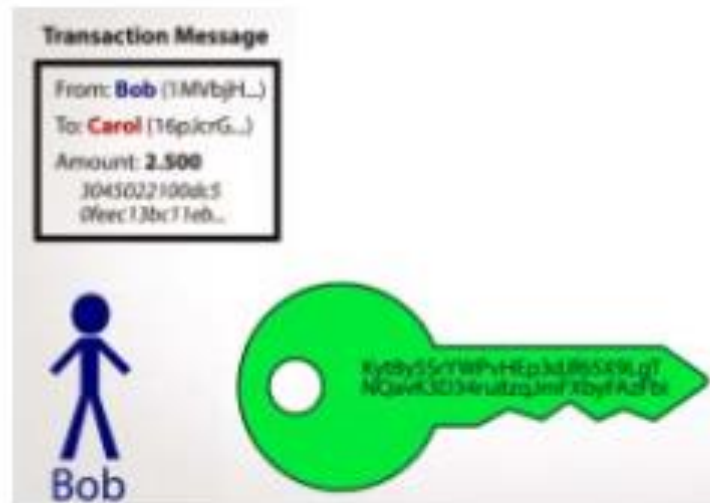
Example- Use eWallet app to submit transaction



Scan recipient's address
and submit transaction



\$ appears in recipient's eWallet

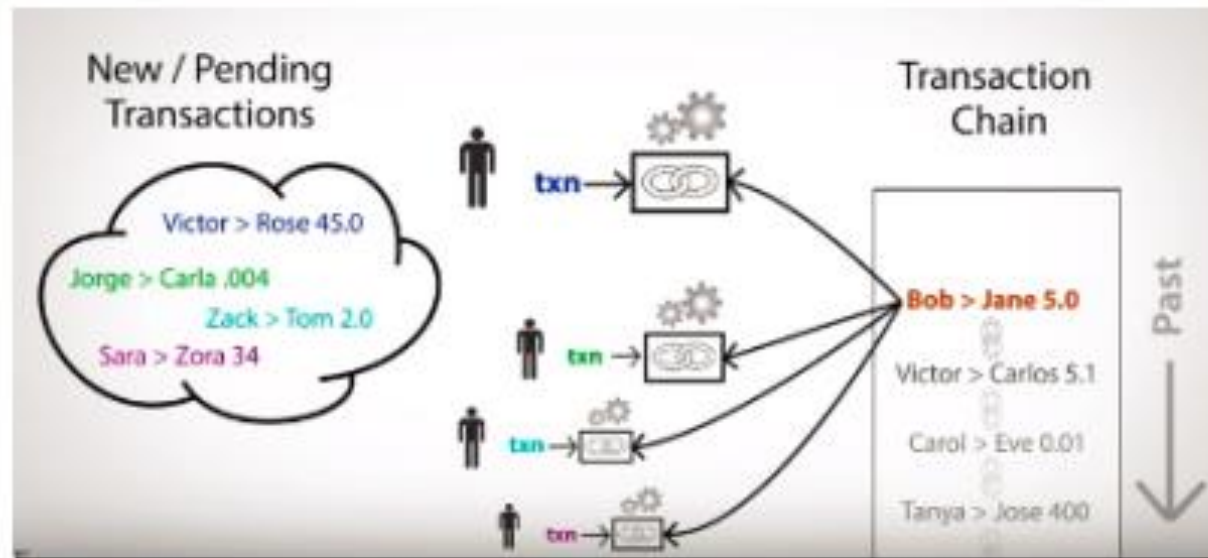


Wallet has keys not money
Creates PKI Signature address pairs

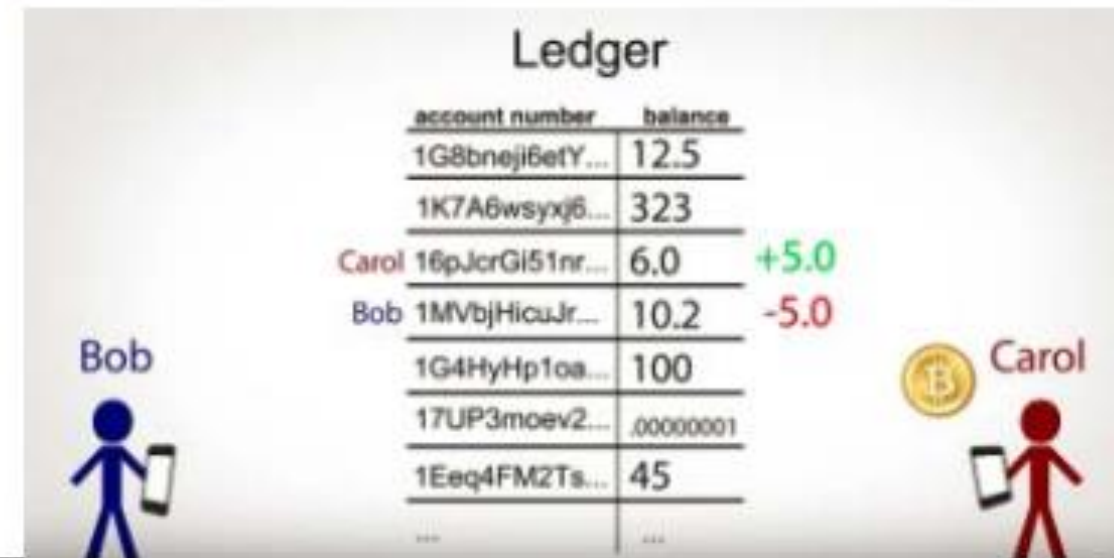


A new PKI hashed signature for each transaction

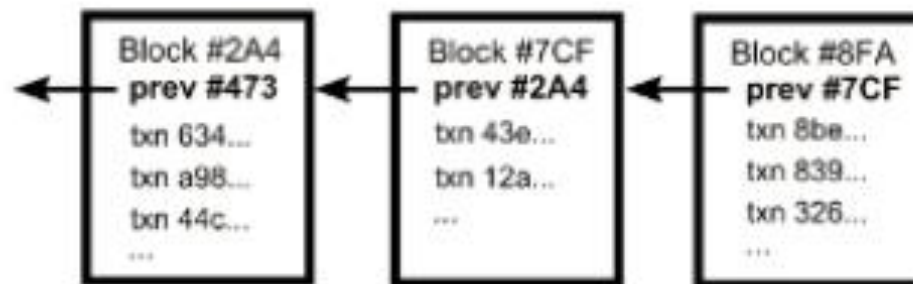
P2P network confirms & records transactions



Transactions submitted to mempool, and miners assemble new batch (block) of transactions each 10 min



Transaction computationally confirmed
Ledger account balances updated



Each block includes a cryptographic hash of the last block, chaining the blocks, hence "Blockchain"



Peer nodes maintain distributed ledger

Key Features of Blockchain Technology

Blockchain technology is built on a foundation of unique characteristics that differentiate it from traditional databases. The following are its most important and defining characteristics:

- **Decentralization**
- **Transparency**
- **Immutability**
- **Tamper-Proof and**
- **Security**

Key Features of Blockchain Technology

Decentralization

- no centralized authority
- operates on a decentralized network of computers (nodes).
- everybody in the network has the copy of distributed ledger
- no one can modify
- **consensus mechanisms** are used to validate and agree on the state of the blockchain

Key Features of Blockchain Technology

Transparency

- The data in a blockchain is visible to all participants in the network.
- Every transaction or record is recorded in a block, and
- the entire history of the blockchain is accessible to anyone who has the necessary permissions.

Key Features of Blockchain Technology

Immutability

- Once a block is added to the blockchain, it is very difficult to alter the information within it.
- Each block contains a **cryptographic hash** of the previous block, creating a chain of blocks.
- Changing the data in one block would require changing the data in all subsequent blocks, which is computationally infeasible.
- Using **Smart Contract Code** for avoiding the alter of agreements. These automate processes and enforce agreements without intermediaries.

Key Features of Blockchain Technology

Tamper-Proof

- Detecting with hashes and blocks.
- Each hash function associated with a block is unique.
- Any change in the data will lead to a change in the hash function.

Security

- Blockchain uses cryptographic techniques to secure transactions and control access to the network.
- Public and private keys enable secure authentication and authorization.
- Consensus mechanisms, such as **Proof of Work (used in Bitcoin)** or **Proof of Stake**, ensure agreement on the state of the blockchain.

Pros and Cons of Blockchain Technology

Pros

- **Transparency.** All participants in a blockchain network share the same documentation instead of individual copies. Because data is shared across a wide network of computers, the blockchain is available for anyone to access, verify and audit data and transactions.
- **Traceability.** Information in a blockchain is organized chronologically and every block of data has an alphanumeric code linked to the preceding block. This makes it easy to track a specific transaction and trace it back to its source.
- **Security.** Data can only be added to a blockchain. Once a block has been created, it cannot be altered, securing the information within it. Moreover, the decentralized nature of most blockchains makes them immune to takeovers or corruption by a central entity.

Pros and Cons of Blockchain Technology

Cons

- **Energy consumption.** The way proof of work networks function is inherently inefficient. Having several network users competing to validate the same operations is a huge waste of energy that has a strikingly negative impact on the environment.
- **Scalability.** As the number of users in a blockchain grows, so does the number of operations. The computational power required for these operations may outpace the workload that hard disks are realistically able to handle.
- **Unavoidable security flaw.** Blockchains are generally more secure than other databases and record-keeping methods but they share one key flaw. If more than half of the nodes participating in the network create invalid transactions or deny legitimate ones, the rest cannot intervene. This is known as a 51% attack.

Distributed Ledger Technology

- Distributed ledger technologies (DLT) were a precursor to blockchain.
- A distributed ledger operates as a ‘network’ in which users approve record of transactions.
- The data is replicated with multiple users and there is no one database.
- Every user or computer on the DLT network with an approved consensus, the ledger is updated with the transaction details.
- There is **no central owner or administrator** of the distributed ledger.
- The data is stored and shared between everyone on the large network irrespective of their location or institution.

Blockchain (Distributed Ledger) vs Traditional Database

WHAT IS BLOCKCHAIN?

Blockchain is a peer-to-peer decentralized distributed ledger technology. It was first introduced in 2009.



WHAT IS A DATABASE?

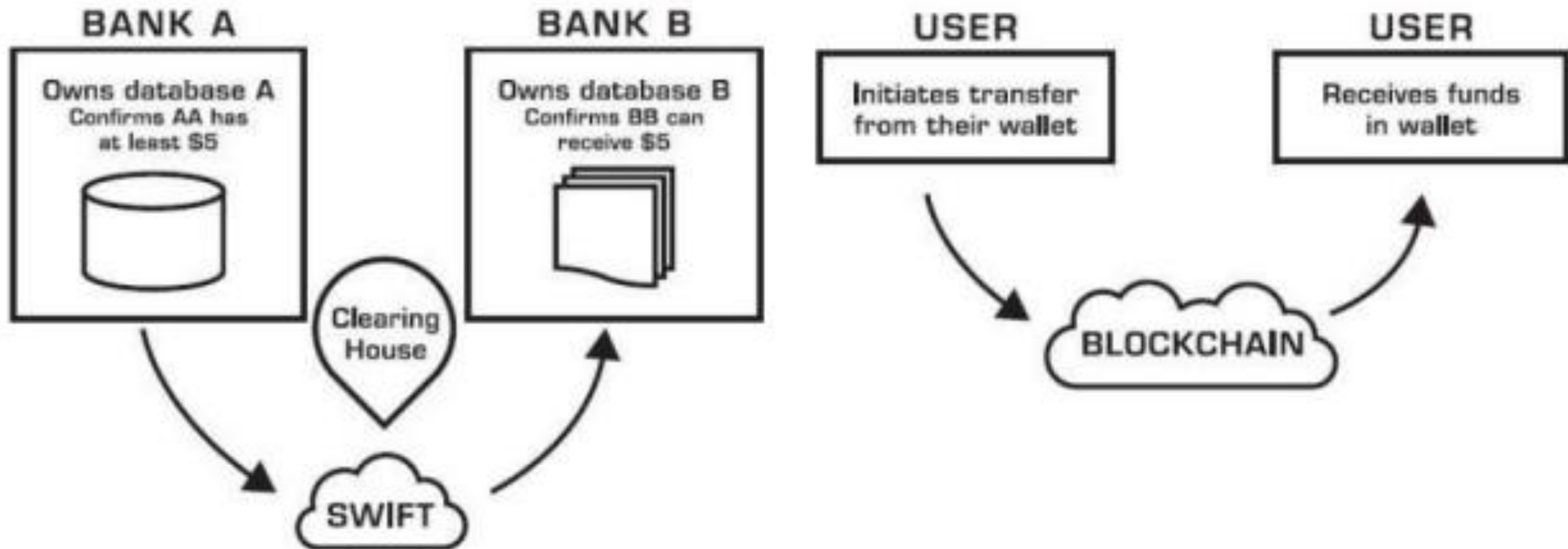
Databases are centralized ledger which stores data in a structured way and is managed by an administrator.



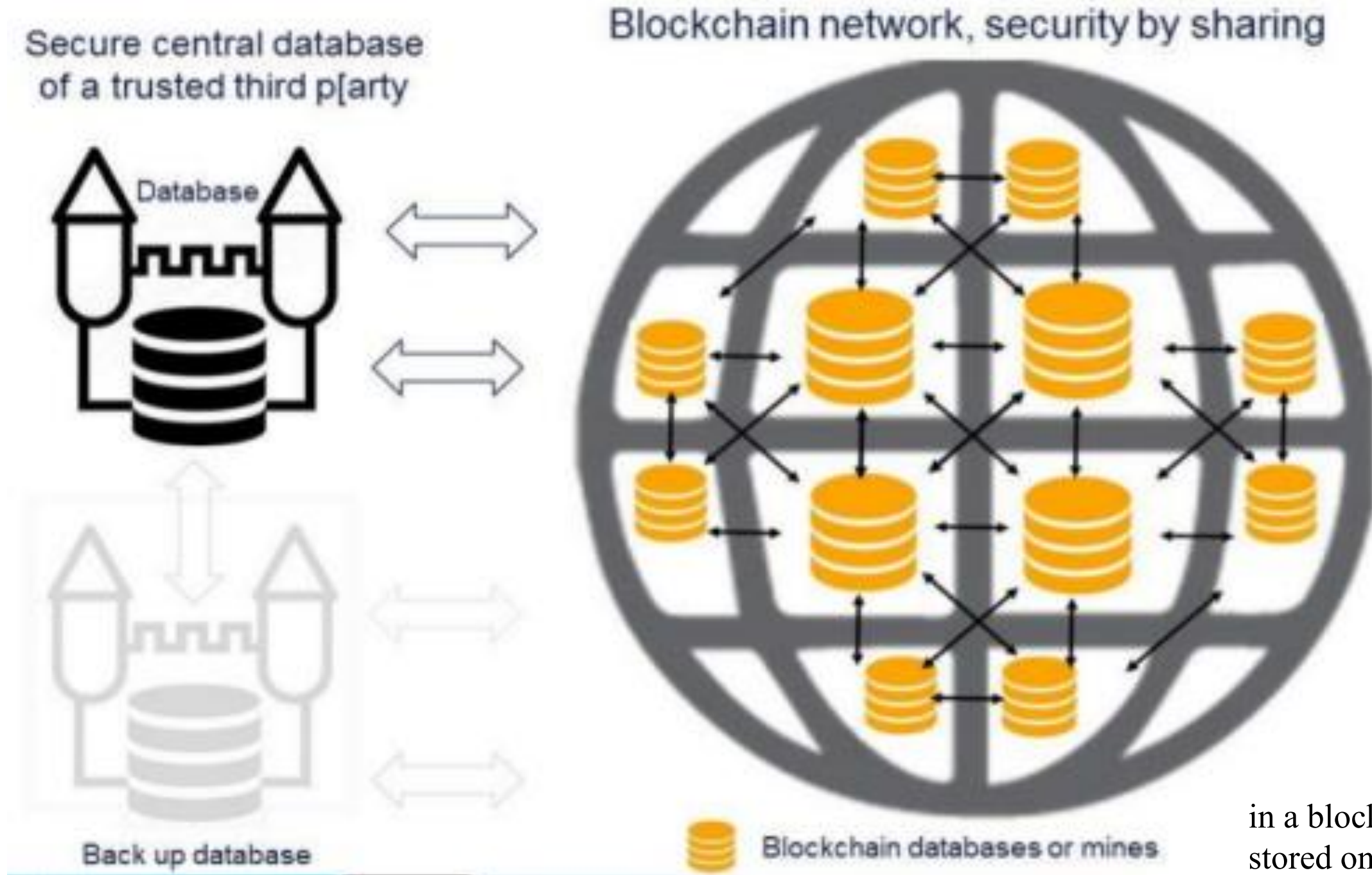
BLOCKCHAIN V/S DATABASE

Blockchain is decentralized and has no centralized approach. However, there are private blockchains that may utilize some form of centralization.	AUTHORITY	Databases are controlled by the administrator and are centralized in nature.
Blockchain uses a distributed ledger network architecture.	ARCHITECTURE	Database utilizes a client-server architecture.
Blockchain utilizes Read and Write operations.	DATA HANDLING	The database supports CRUD (Create, Read, Update and Delete).
Blockchain data supports integrity.	INTEGRITY	Malicious actors can alter database data.
Public blockchain offers transparency.	TRANSPARENCY	Databases are not transparent. Only the administrator decides which the public can access data.
Blockchains are comparatively harder to implement and maintain.	COST	The database being an old technology is easy to implement and maintain.
Blockchain is bobbed down by the verification and consensus methods.	PERFORMANCE	Databases are extremely fast and offer great scalability.

Blockchain (Distributed Ledger) vs Traditional Database



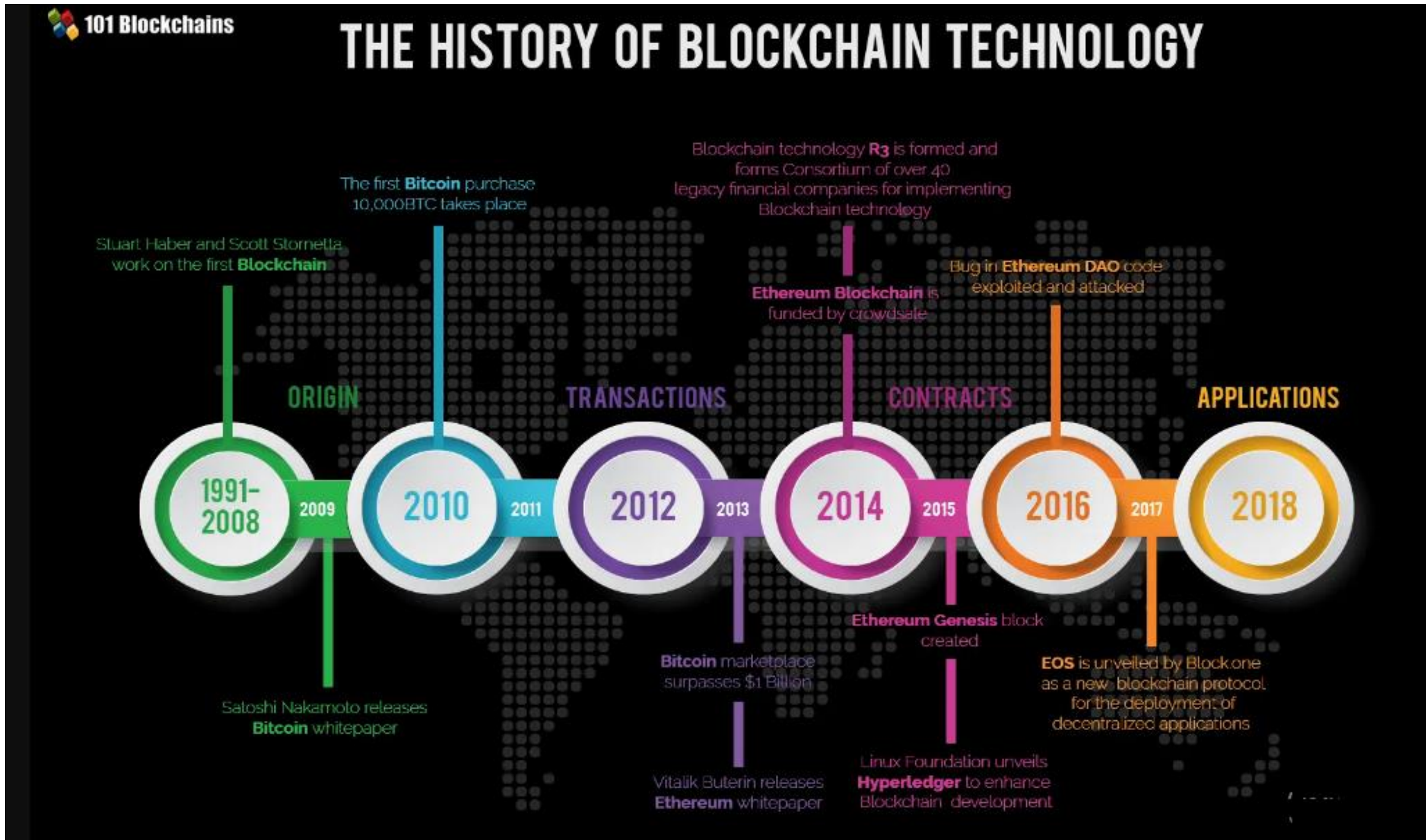
Blockchain (Distributed Ledger) vs Traditional Database



in a blockchain network the data is stored on many computers (the so-called '[miners](#)')

History and Evolution of Blockchain Technology

History of Blockchain Technology



History of Blockchain Technology

1991-2008: Early Years of Blockchain Technology

How did blockchain emerge?

Stuart Haber and W. Scott Stornetta envisioned what many people have come to know as blockchain, in 1991. Their first work involved **working on a cryptographically secured chain of blocks** whereby no one could tamper with the timestamps of documents.

In 1992, they upgraded their system to **incorporate Merkle trees that enhanced efficiency** thereby enabling the collection of more documents on a single block. However, it is in 2008 that Blockchain History starts to gain relevance, thanks to the work of one person or group by the name of Satoshi Nakamoto.

History of Blockchain Technology

Satoshi Nakamoto is accredited as the brains behind blockchain technology. Very little is known about Nakamoto as people believe he could be a person or a group of people that [worked on Bitcoin](#), the first application of [digital ledger technology](#).

Nakamoto conceptualized the [first blockchain in 2008](#) from where the technology has evolved and found its way into [many applications beyond cryptocurrencies](#). Satoshi Nakamoto released the first whitepaper about the technology in 2009. In the whitepaper, he provided details of how the technology was well equipped to [enhance digital trust given the decentralization](#) aspect which meant nobody would ever be in control of anything.

Ever since Satoshi Nakamoto exited the scene and handed over Bitcoin development to other core developers, the digital ledger technology has evolved resulting in new applications that make up the blockchain History.

Evolution of Blockchain Technology

Phase 1- Transactions

2008-2013: Blockchain 1.0: Bitcoin Emergence

- The 1st generation of the blockchain protocol, created in 2009, and related to the Bitcoin protocol introduced by Satoshi Nakamoto
- Implementation of DTL that enables the use of **currency transaction (cryptocurrencies)** among individuals

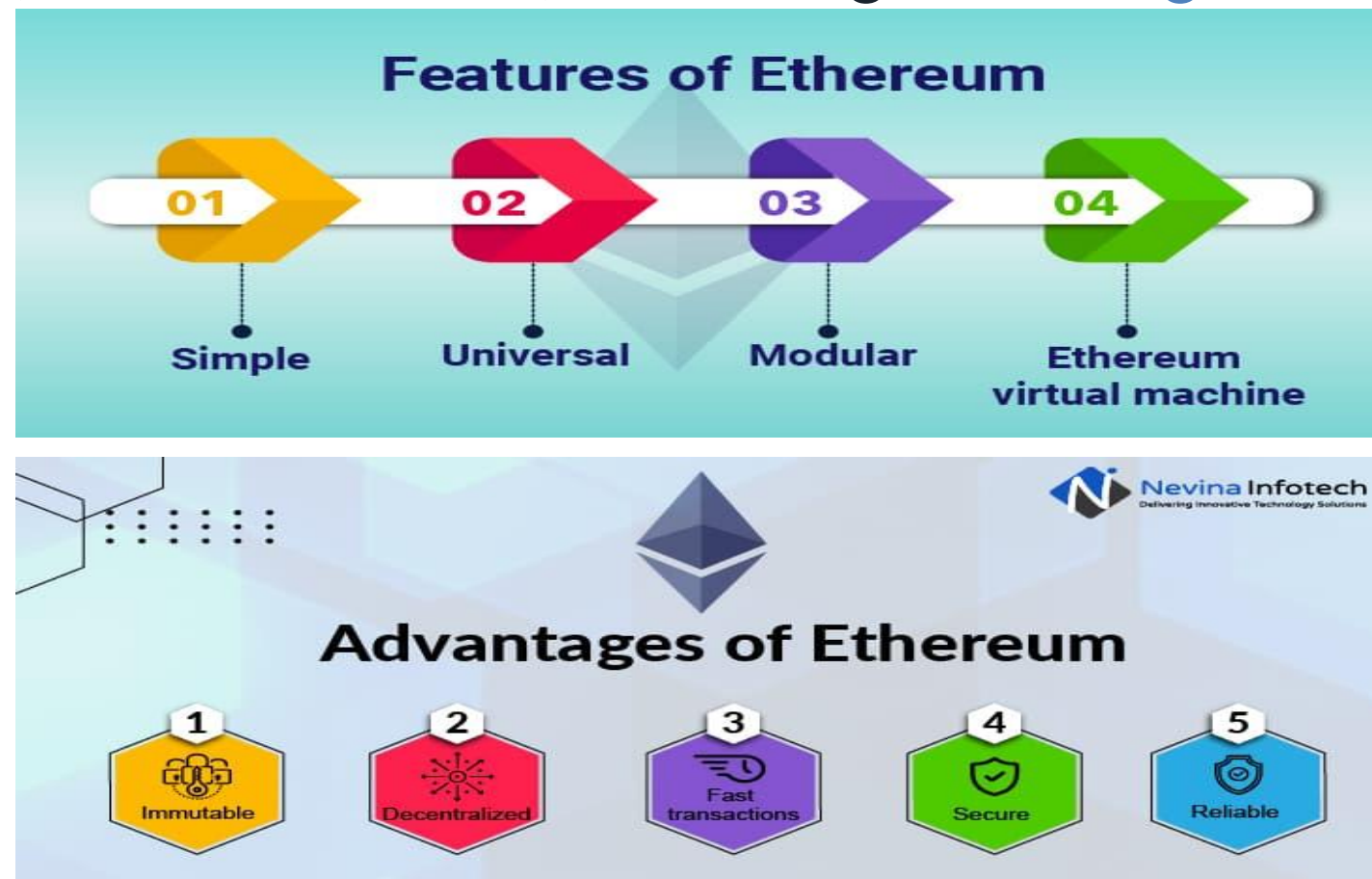


Evolution of Blockchain Technology

Phase 2- Smart Contracts

2013-2015: Blockchain 2.0: Ethereum Development

- New applications of the blockchain protocol **outside the monetary transactions and the financial system**
- Applies in the **Ethereum protocol**, created in 2013, that brings the **usage of smart contracts within the blockchain**

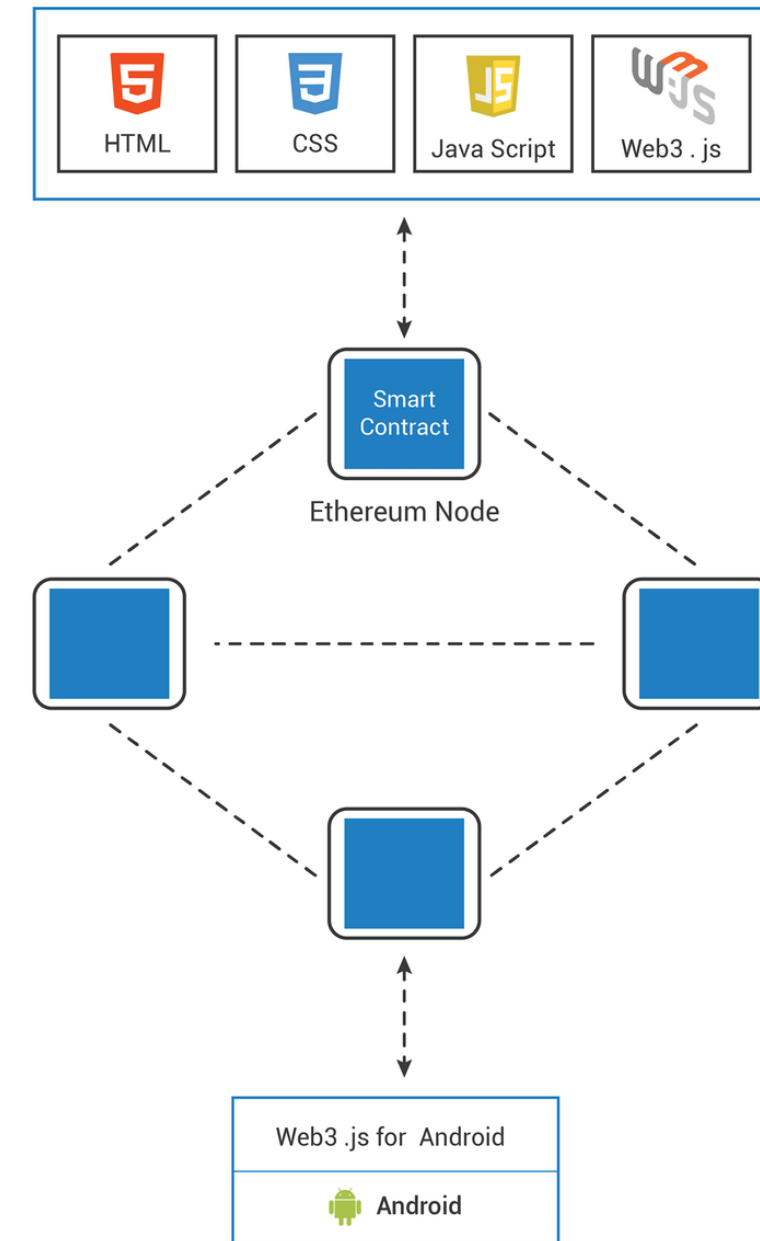


Evolution of Blockchain Technology

Phase 3- Decentralized Applications (DApps)

2018-2021: Blockchain 3.0: the Future

- Implemented in 2017, constitutes the use of decentralized applications to store and communicate between entities, serving as a frontend application to users, with backend calls to several procedures
- Improves the blockchain protocol for a faster consensus and quicker transaction confirmation



https://www.researchgate.net/figure/The-architecture-of-decentralized-applications-Dapp_fig6_342947289

Evolution of Blockchain Technology

Phase 3- Decentralized Applications (DApps)

2018-2021: Blockchain 3.0: the Future



https://medium.com/@may_sanders/dapp-development-guide-7-steps-to-streamline-your-development-process-993d3ffa5cb5

Evolution of Blockchain Technology

Smart Contracts and Decentralized Applications

Smart Contracts and Decentralized Applications



<https://fastercapital.com/topics/smart-contracts-and-decentralized-applications.html>

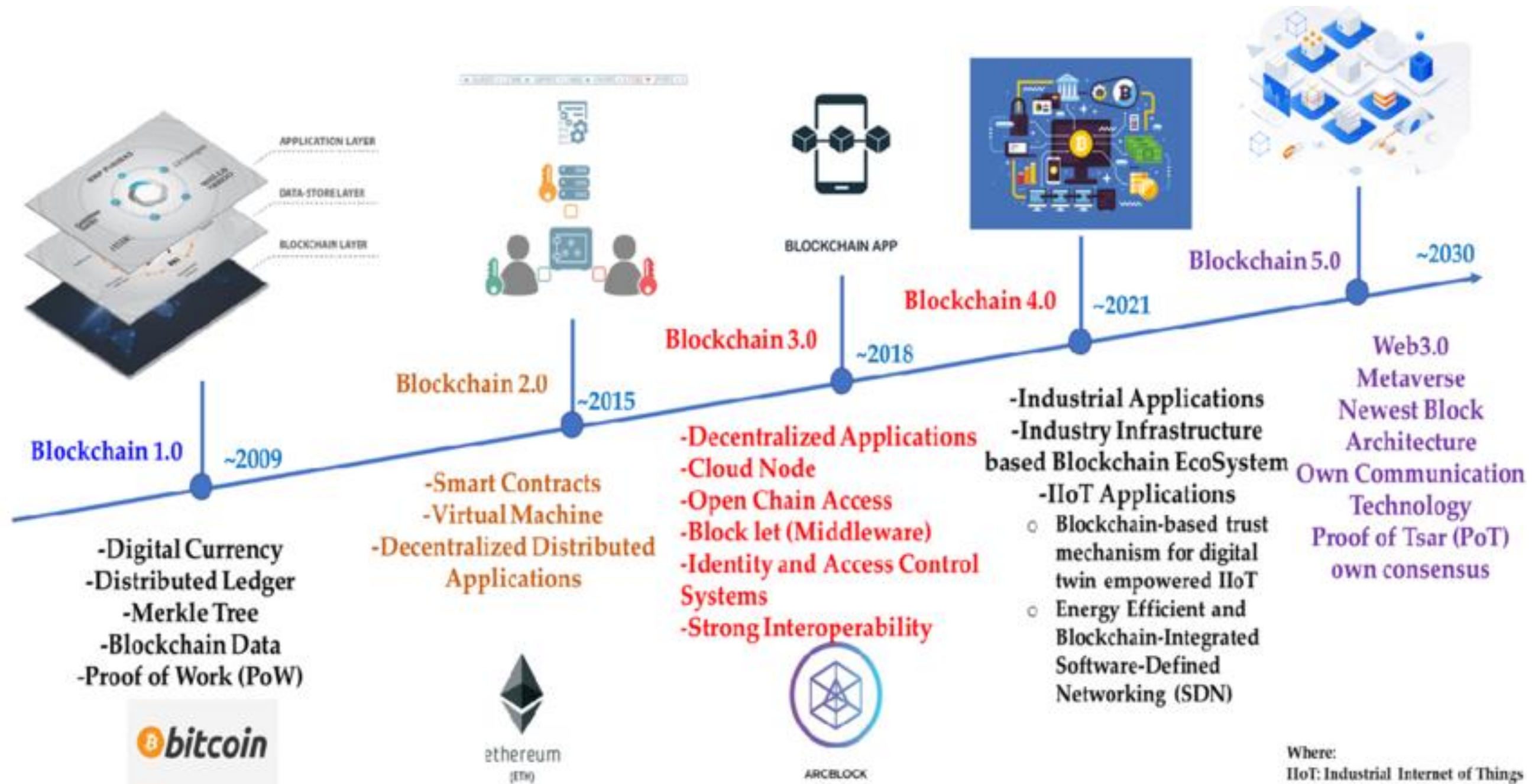
Evolution of Blockchain Technology

Phase 4 - Industry Usage

2021: Blockchain 4.0: the Future

- A 4th generation that is in testing/development, to improve the blockchain network into finding a place in the **industry areas**
- Promises to bring all the required functions to **create a fully automated and integrated system** in the blockchain network for the industrial demands
- When completely distributed, this version is going to revolutionize the blockchain protocol, in the **appearance of new areas of application**

Evolution of Blockchain Technology



https://www.researchgate.net/figure/Evolution-from-Blockchain-10-up-to-Blockchain-50_fig3_367324333

Types of Blockchain

Popular Blockchain Platforms

- The blockchain is a digital ledger of economic transactions, which programmed across a peer-to-peer network.
- Using the Blockchain Technology, participants can confirm transactions without any need for a central clearing authority.
- Many Blockchain Consulting companies are currently active using some of the Blockchain Platforms like-
 - **Bitcoin**
 - **Ethereum**
 - **Hyperledger and**
 - **Quorum**

What is Bitcoin?

Bitcoin is cryptocurrency a digital asset that can be sent anywhere in the world over the Internet.

When did it start?

It was launch in 2009 by **Satoshi Nakamoto**.

How does it work?

Bitcoin works via **open sources** software over a **peer- to- peer network**, by passing banks and traditional payment gateways. No central authority needed.

What is Bitcoin Fork?

A Bitcoin Fork is defined as a blockchain split into two paths forward, or a **change of protocol rules**.



Is bitcoin Safe?

A bitcoin transfer is **safer than credit card transactions** as it doesn't involve personal and credit card data.

Who uses Bitcoin and who accepts it?

Bitcoin seems to be used by people who easily have access to the internet and formal financial services in general. A limited number of Businesses accept **bitcoins as payment for good or services**. i.e. Amazon, okCupid, western union.

Some Sources of Bitcoin to learn.

bitcoin.org

coindesk.com

buyphysicalbitcoins.com

Ethereum

What is Ethereum?

Ethereum is an **open-source**, decentralized blockchain-based platform, for featuring **smart contract functionality**.

Different platforms of Ethereum

Virtual Machine
Smart Contracts
Applications
Performance

What will Ethereum be used for?

Decentralizing Existing services.
Bringing Science Fiction to Life.
For Unimagined Possibilities.

What is Smart Contract?

Smart contract is a **computerized transaction protocol** that help us exchange of money, content, property, shares, or anything of value



Which apps currently runs on Ethereum?

Etheria
Ether Tweet
Raiden Network
TenX
Gnosis

Which apps currently runs on Ethereum?

Immutability
Ethereum is Secure
It has Zero Downtime
Corruption and tamper proof

Coming Milestones.

Scalability solutions such as sharing and Plasma promise to bring ETH to 1000s of transactions per second

Want more Info?

Ethereum.org
Stateofthedapps.com
Coinlib.io
Entethalliance.org

Hyperledger

What is HyperLedger?

Hyperledger is an **open source** collaborative effort created to advance cross-industry blockchain technologies, It was hosted in **December 2015** by **Linux** Foundation and supported by big industry players like IBM, Intel.

Hyperledger Frameworks.

Hyperledger Burrow.
Hyperledger Fabric.
Hyperledger Iroha.
Hyperledger Sawtooth.
Hyperledger Indy.

Hyperledger Tools.

Hyperledger Caliper.
Hyperledger Cello.
Hyperledger Composer.
Hyperledger Explorer.
Hyperledger Quilt

History of Hyper ledger

The objective of this project is to **collaborate with cross industry** by developing blockchains and distributed ledgers, with a particular focus on improving **the reliability and performances of these systems** so that they're capable of supporting world business transactions by major technological, financial and supply chain companies.



Hyperledger Projects

The Hyperledger incubates and promotes a range of Business.
Hyperledger Sawtooth.
Hyperledger Iroha.
Hyperledger Burrow
Hyperledger Fabric and many more.

Source

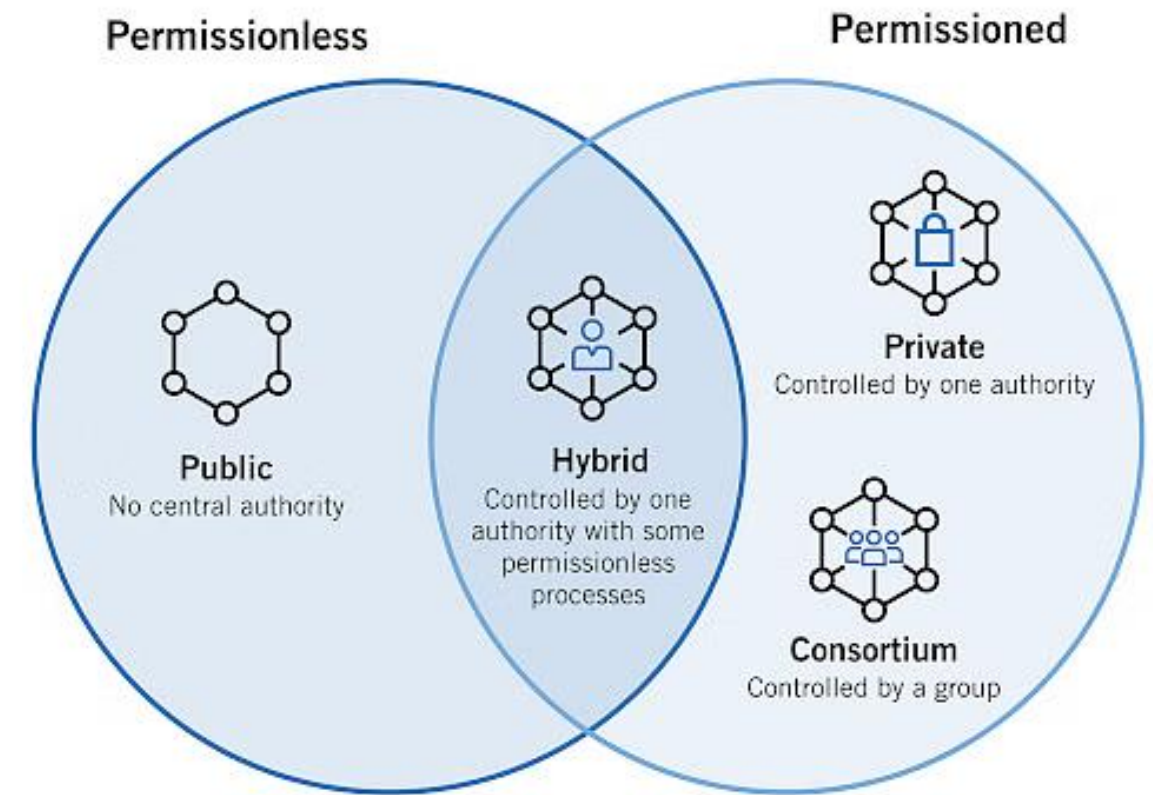
[Hyperledger.org](https://hyperledger.org)
[Blockgeeks.com](https://blockgeeks.com)

How do Bitcoin, Ethereum, and Hyperledger Fabric differ?

Blockchain characteristics comparison			
Characteristics	Bitcoin	Ethereum	Hyperledger
Permission restrictions	Permissionless	Permissionless	Permissioned
Restricted public access to data	Public	Public or private	Private
Consensus	Proof-of-Work	Proof-of-Work	PBFT
Scalability	High node-scalability, Low performance-scalability	High node-scalability, Low performance-scalability	Low node-scalability, High performance-scalability
Centralized regulation (governance*)	Low, decentralized decision making by community/miners	Medium, core developer group, but EIP process	Low, open-governance model based on Linux model
Anonymity	Pseudonymity, no encryption of transaction data	Pseudonymity, no encryption of transaction data	Pseudonymity, encryption of transaction data
Native currency	Yes, bitcoin, high value	Yes, ether	No
Scripting	Limited possibility, stack-based scripting	High possibility, Turing-complete virtual machine, high-level language support (Solidity)	High possibility, Turing-complete scripting of chaincode, high-level Go-language

Types of Blockchain Technology

- Blockchain technology has evolved into a versatile tool with various applications across industries.
- Understanding the **different types of blockchain** is essential for **selecting the right solution** for specific needs.
- Broadly categorized into public, private, consortium, and hybrid blockchains, each type offers unique characteristics, benefits, and use cases.
- **Public blockchains** enable open access and decentralization, while **private blockchains** prioritize security and control.
- **Consortium blockchains** serve collaborative networks, and **hybrid blockchains** combine features of both public and private models.



https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.simplilearn.com%2Ftutorials%2Fblockchain-tutorial%2Ftypes-of-blockchain&psig=AOvVaw3dVinyEqAbfV_xHO4nG-J&ust=1731920067291000&source=images&cd=vfe&opi=89978449&ved=2ahUKEwiVlbKU_-KJAxWxcWwGHfO2CecQjRx6BAGAEbk

Types of Blockchain Technology

Permissionless Blockchain

- A permissionless blockchain is a type of blockchain network that **allows anyone to participate in the network** without requiring special permissions or approvals.

Permissioned Blockchain

- A permissioned blockchain is a type of blockchain network that **restricts access** and participation to a select group of authorized users.
- Unlike permissionless blockchains, where anyone can join and validate transactions, permissioned **blockchains require participants to obtain permission** before they can access the network or perform certain actions.

Types of Blockchain Network

Public Blockchains

- They are **open to everyone**, and all transactions on the network can be traced.
- Public blockchains are also known as **permissionless blockchain** network because anyone who **joins can read or write** to them anonymously without the need for authorization.

Private Blockchains

- Private blockchains are fundamentally different from public blockchains because they're run by a **central authority** – they're **not entirely decentralized networks**.
- An **industry group, or a group of companies** owned by an umbrella company.
- The central authority determines who can read, write and participate in the network's activities, which is why they're also known as **permissioned blockchain** networks.
- Private blockchains offer **greater customizability** and can be used to **store sensitive data**.



Types of Blockchain Network

Consortium Blockchains

- Consortium blockchains are most similar to private networks but share some characteristics with public blockchains. These are controlled by multiple central authorities and collaborate on a decentralized network. A predefined group of individuals or nodes are in charge of reaching consensus in consortium blockchains. Like hybrid blockchains, the ability to read or write on the network may be public or restricted to select participants.

Hybrid Blockchains

- Hybrid or semi-private blockchains were built to offer the best of both worlds by combining elements from both public and private types of blockchain. For example, they're run by a single entity but may employ both permission-based and permissionless systems to fine-tune access to their data. Transactions are generally not public but may be verified if necessary. Users join the network as anonymous participants and only reveal their identities to other parties when they engage in transactions.

Comparative Analysis of Blockchain Types

Feature	Public Blockchain	Private Blockchain	Hybrid Blockchain	Consortium Blockchain
Access Control	Open to everyone	Restricted to specific participants	Limited to a group of organizations	Combination of public and private
Governance	Decentralized	Centralized	Semi-decentralized	Mixed governance structure
Transparency	High transparency	Low transparency	Moderate transparency	Variable transparency
Scalability	Limited scalability	High scalability	Moderate scalability	High scalability potential
Security	High due to decentralization	Lower due to centralization	Moderate security	Variable security
Transaction Speed	Slower due to consensus mechanisms	Faster transactions	Faster than public, slower than private	Variable speed
Use Cases	Cryptocurrencies, decentralized apps	Enterprise solutions, data privacy	Supply chain, banking, collaborations	Various applications need flexibility

Technical Concepts of Blockchain

Elements of Blockchain

- Nodes
- Miners
- Ledger (Block/Blockchain)
- Cryptographic Algorithm
- Consensus Mechanism
- Smart Contract
- Double spending problem

Elements of Blockchain

Node

- A node in a blockchain can be **any electronic device** that is part of a peer-to-peer network and maintains its copy of the blockchain.

Mining

- It is the process of **verifying and recording new transactions** on the blockchain performed by Miners, for which they have **special mining software**.

Ledger

- The ledger **records transactions** such as payment, supply chain details, medical, real estate contracts, etc.

Consensus

- Nodes within a distributed blockchain network must agree on the actual state of the network to ensure the **validity of transactions using Proof of Work (PoW) or Proof of Stake (PoS)**.

Cryptographic Algorithm

- **SHA-256** is a cryptographic algorithm that accepts input of any length, deterministically encodes the data, and returns a **hash of 256 bits or 64 characters**. The SHA-256 hashing algorithm ensures that the output can never track back to the input, making it very secure.

Smart Contract

- Smart contracts are **digital contracts** stored on a blockchain that are automatically executed when **predetermined terms and conditions** are met.

Double-Spending Problem

- double spending happens when someone tries to use the **same digital money more than once**.

What is Double Spending in Blockchain?

- In the world of digital currency, double spending is a big concern.
- In blockchain technology, double spending happens when someone tries to use the same digital money more than once.
- Blockchains are designed to prevent this by keeping a public record of all transactions.
- Every time someone spends cryptocurrency, the blockchain updates to show that the money has already been used.
- This way, it's almost impossible to cheat and spend the same digital money more than once.
- Blockchain use distributed ledger technology (DLT). Every user or computer on the DLT network with an **approved consensus**, the ledger is updated with the transaction details.

What is Double Spending in Blockchain?

Although Blockchain is secured, still it has some loopholes. Hackers or malicious users take advantage of these loopholes to perform their activities.

1. Double spending means the expenditure of the same digital currency twice or more to avail the multiple services. It is a technical flaw that allows users to duplicate money.
2. Since digital currencies are nothing but files, a malicious user can create multiple copies of the same currency file and can use it in multiple places.
3. This issue can also occur if there is an alteration in the network or copies of the currency are only used and not the original one.
4. There are also double spends that allow hackers to reverse transactions so that transaction happens two times.
5. By doing this, the user loses money two times one for the fake block created by the hacker and for the original block as well.
6. The hacker gets incentives as well for the fake blocks that have been mined and confirmed.

The Double Spending Problem -Example

Double spending occurs when someone tries to use the same digital currency more than once. Here's a simplified example using transactions to illustrate how it can happen:

- 1. Initial Transaction:** Alice has 1 Bitcoin (BTC). She decides to send 1 BTC to Bob. This transaction is broadcast to the network and is pending confirmation.
- 2. Attempt to Double Spend:** While the transaction to Bob is still pending, Alice tries to spend the same 1 BTC again. She creates a second transaction, sending the same 1 BTC to Charlie. This second transaction is also broadcast to the network.
- 3. Network Propagation:** The two transactions (to Bob and Charlie) are now competing to be included in the blockchain. Each transaction is seen by different nodes in the network, and there may be a delay in the entire network agreeing on which transaction is valid.
- 4. Block Confirmation:** The network eventually confirms one of the transactions (e.g., the transaction to Bob). This transaction is added to a block in the blockchain, making it permanent.
- 5. Conflict Resolution:** If the transaction to Bob is confirmed first, the network sees that Alice has already spent the 1 BTC, so the transaction to Charlie is invalidated. If the transaction to Charlie somehow gets confirmed first, Bob's transaction will be rejected.
- 6. Final Outcome:** The blockchain only records one transaction because it is designed to prevent double-spending. The system resolves the conflict by ensuring that only one of Alice's transactions is valid.

Technical Concepts of Blockchain Technology

- Blockchain combines existing technologies to prevent the double-spend problem.
 - Distributed Systems
 - Peer-to-peer networks
 - Hashing functions
 - Public - Private key cryptography
 - Cryptographic signatures and
 - Consensus
- blockchain uses the three principles of cryptography, decentralization, and consensus to create a highly secure underlying software system that is nearly impossible to tamper with. There is no single point of failure, and a single user cannot change the transaction records.

Cryptography and Decentralization

Cryptography

- In blockchain, cryptography is used for the following two purposes:
 1. **Securing the identity** of the sender of transactions
 2. Ensuring that past **records cannot be tampered** with

Decentralization

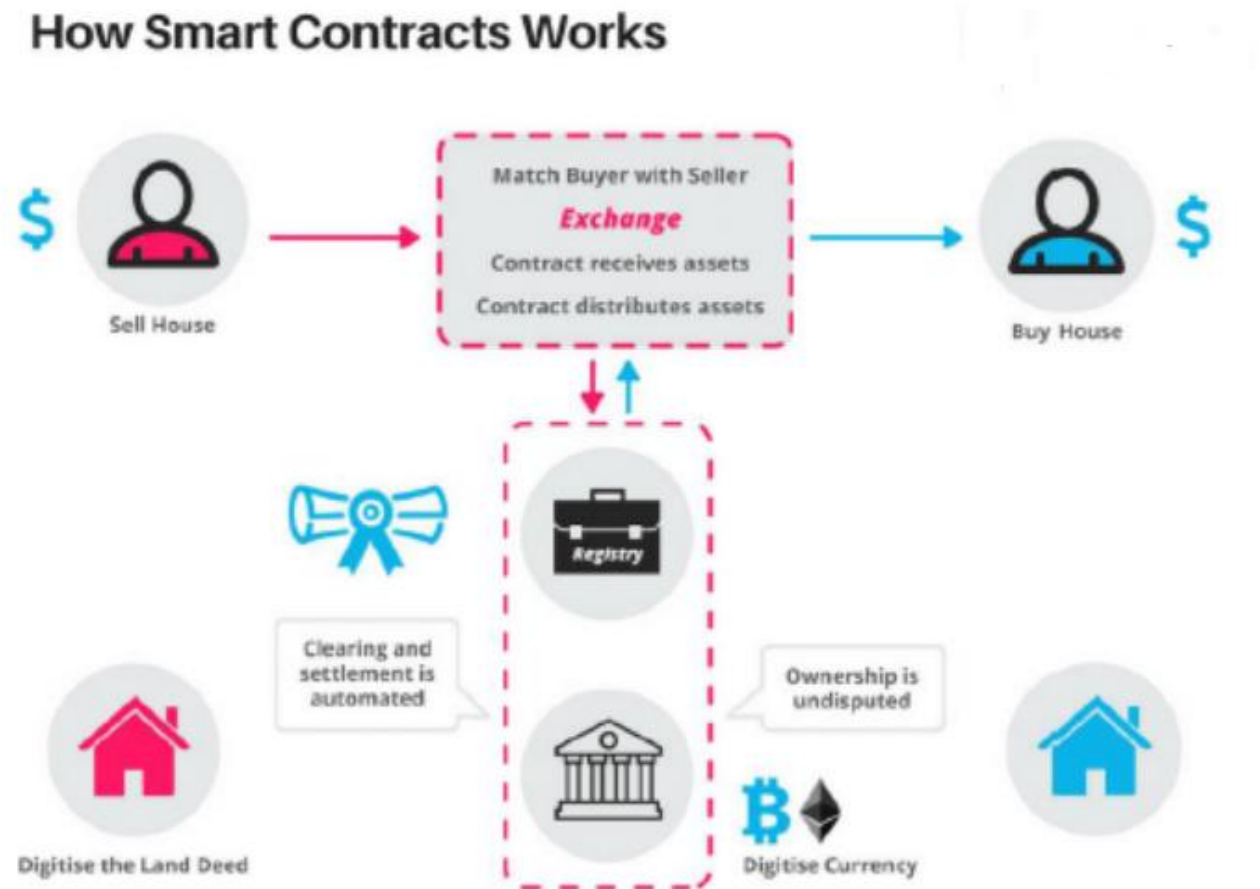
- In blockchain, decentralization refers to the **transfer of control and decision-making** from a centralized entity (individual, organization, or group thereof) to a distributed network

Consensus

- Blockchains are peer-to-peer networks with no central administrator or authority.
- It is crucial to ensure that the network participants reach consensus on the state of the ledger i.e. the **uniqueness and order of records**.
- This is done through **consensus algorithms that apply different methods** to ensure that the right order and uniqueness of transactions has been determined and validated by enough users to be added to the ledger.
- Consensus Methods
 - 1) Proof of Work (PoW)
 - 2) Proof of Stake (PoS)
 - 3) Simplified Byzantine Fault Tolerance (SBFT)

Smart Contract

- a computer program that is capable of running a set of predefined functions when a specified condition or set of conditions occurs.
- addresses issues that require agreements with minimal trust between the parties involved in a distributed system.



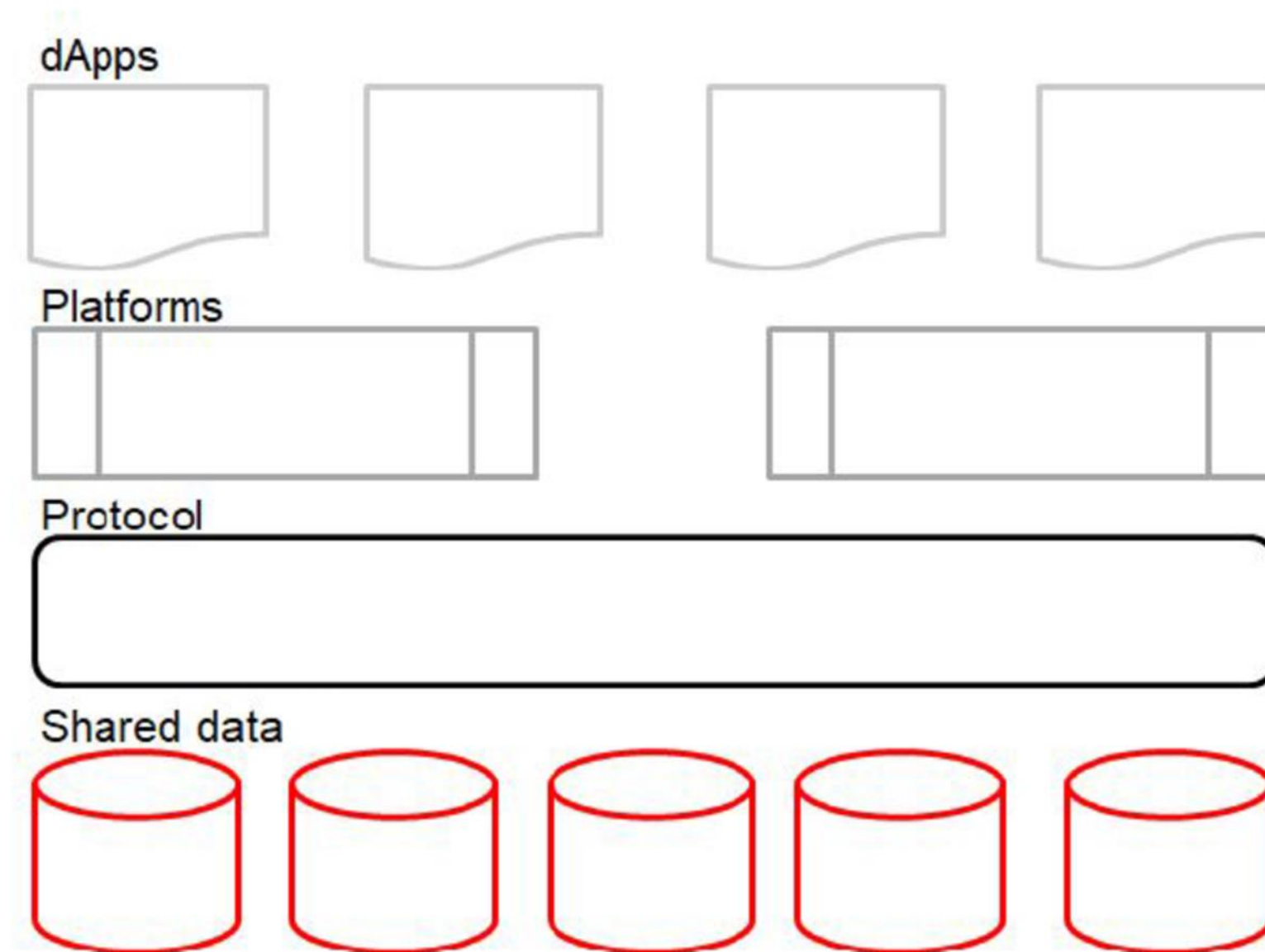
Blockchain Architecture and Use Cases

Blockchain Technology Stack

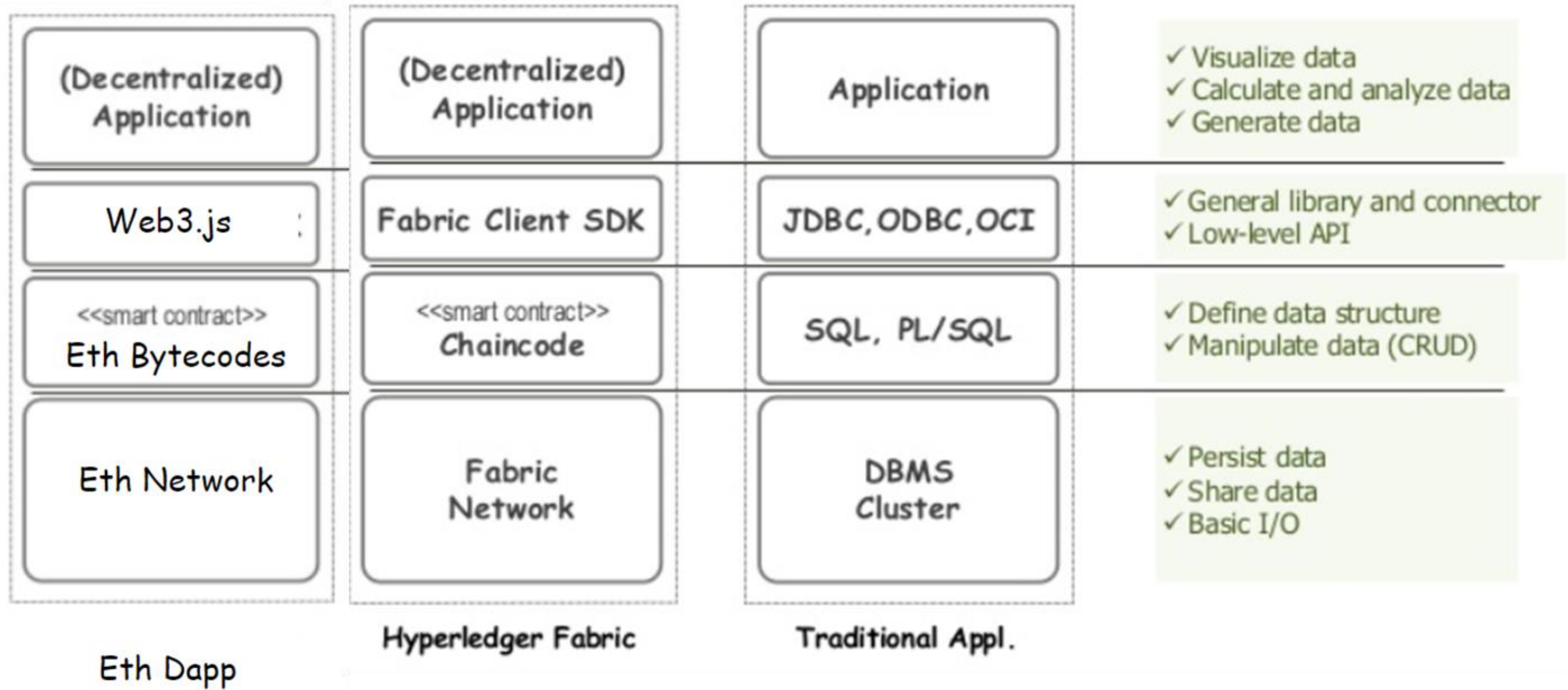
The Blockchain technology stack can be viewed as **four layers** of components:

- 1) Shared Data
- 2) Protocol
- 3) Platforms
- 4) Products / dApps / Smart Contracts

Layers of Blockchain



Layers of Blockchain



Layers of Blockchain

Shared Data

- This is the **decentralized database** that stores all the **transactions in hashed format**.

Protocols

- Examples of existing protocol infrastructure on the web today include TCP/IP, SMTP, HTTP, and HTTPS. Protocols are essentially the infrastructure that everyone who is part of the blockchain ecosystem.
- Blockchain protocols **implement rules for consensus, validation, incentive, and participation**.
- **Bitcoin and Ethereum are examples of protocols**. Bitcoin, being the first and largest example, has protocols in place **to prevent a "double-spend" attack**, allow for peer-to-peer payments, and ensure a strong and verified settlement layer.
- Other types of protocols incorporate other aspects to allow for more features and address different problem sets.

Layers of Blockchain

Platforms

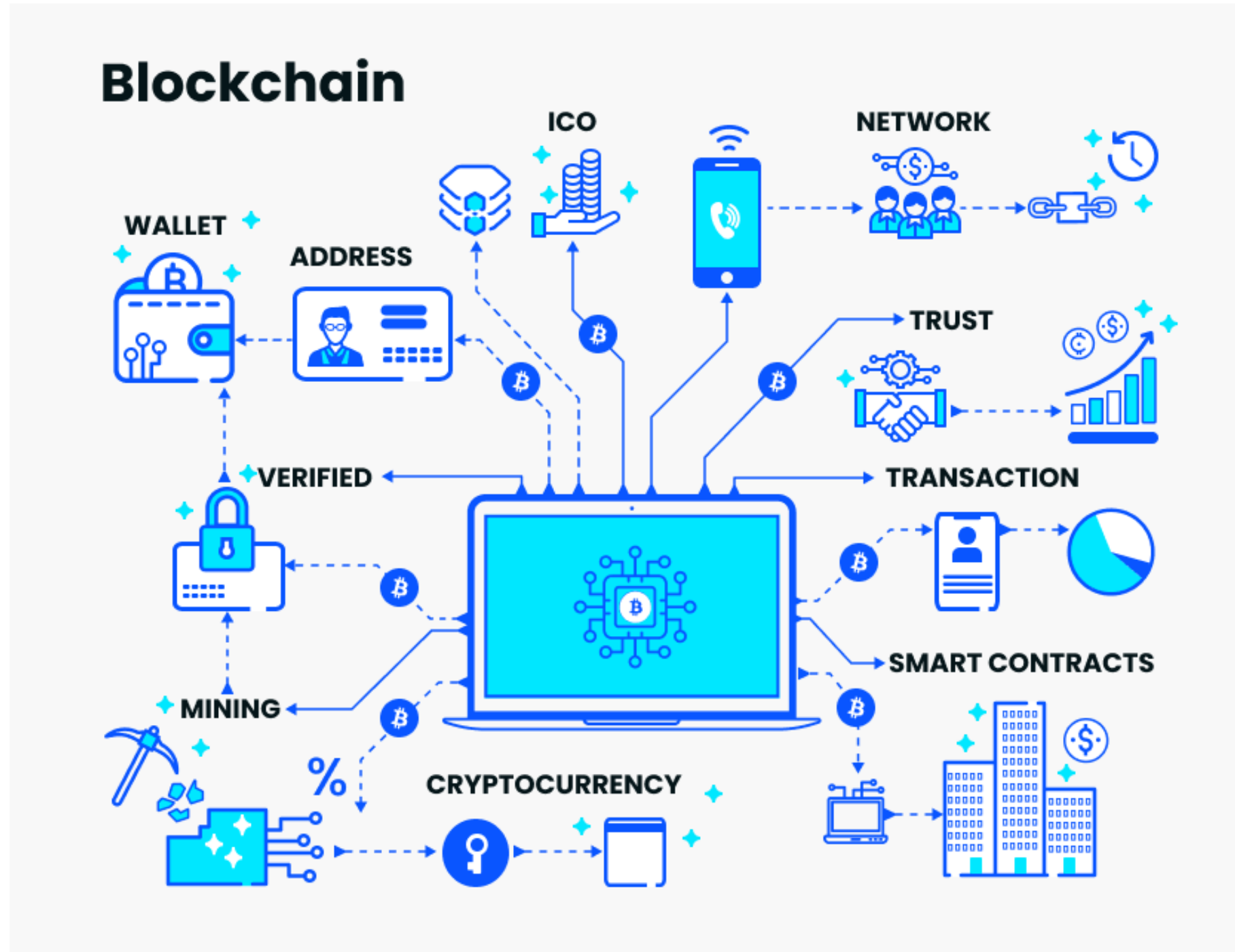
- Platforms are a kind of [middleware](#).
- They allow developers to [build applications on top of protocols](#). Blockchain platforms (Blockchain 2.0) take the concepts introduced by Bitcoin's blockchain protocol and try to expand it to become universal and “turing complete”.
- Platforms seek to act as a “universal computer” that allow the development of applications on top of their protocol layer. Examples of known blockchain platforms include [Ethereum, NEO, and EOS](#).
- Each utilize the technology pioneered by Bitcoin and aim to expand on the [protocol by incorporating "smart contracts."](#)
- This new advancement in the blockchain protocol allows for the use of universal functions to be built on the blockchain infrastructure.

Layers of Blockchain

Products

- Products are the interface to the protocols and platforms.
- They allow users to interact with the protocol and shared data.
- Developers use platforms to build products. Examples of products are dApps (Decentralized Applications).
- These "dApps" utilize blockchain technology alongside platform capabilities like "smart contracts" to provide not only the security of the blockchain, but also the ability to self execute operations.
- "dApps" are the thin application layer built on top of a blockchain protocol layer and allow for trustless, peer-to-peer, decentralized applications.

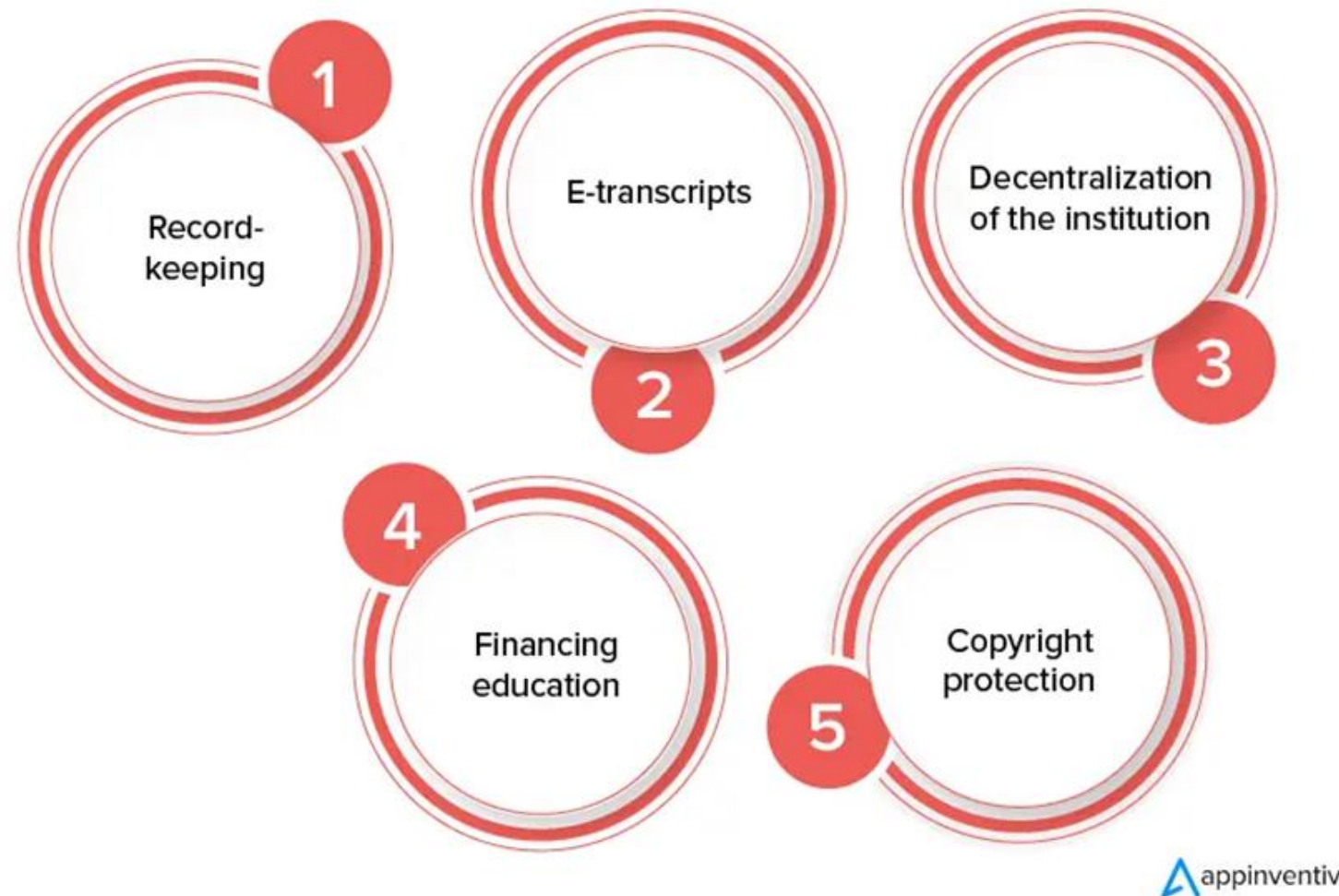
Where Can We Use Blockchain Technology?



Blockchain Use cases

- Education
- Supply Chain Management
- Digital Identity
- Voting
- Healthcare
- Notary
- Food Safety
- Intellectual Property (IP)
- Energy Market
- Real Estate

Key Use Cases of Blockchain in Education



the blockchain in education are: consulting **certificates** or **qualifications**, handling student records, managing intellectual property and payments.

Supply Chain ,Healthcare and Identity Management

Supply chain management: The end-to-end visibility, traceability and accountability of blockchain is useful in managing supply chains. Stakeholders can record, track and authenticate products, prevent counterfeit goods from getting into the supply chain, and streamline logistics processes.

Healthcare: Along with artificial intelligence and IoT, blockchain has emerged as an innovative healthcare technology. In healthcare, blockchain is used to securely store and share patient data. The technology lets patients control their medical records, granting access to healthcare providers only when necessary. This enables seamless and secure sharing of medical information, improving treatment outcomes and reducing administrative burdens.

Identity management: Blockchain-based identity management systems enhance security, privacy and control over personal data. By storing identity information on the blockchain, users can have a portable and verifiable digital identity. This eliminates the need for multiple identity documents, reduces identity theft and simplifies identity verification processes.

Food Safety

Walmart works with IBM to improve the tracking and traceability of food products, leading to better food safety. When integrated with the latest data collection technologies, blockchain has huge potential in the food industry. Based on the respective strengths of blockchain and the Internet of Things (IoT), we can revolutionize the food industry. IoT solutions connect the physical and digital worlds, capturing data such as temperature and humidity during product transport or storage. Blockchain provides a secure and immutable platform where all participants in the supply chain can store and access this data.

Some of the advantages are:

It will streamline the supply chain and reduce costs for retailers

It offers easier compliance

It will improve and speed up the process of downloading food

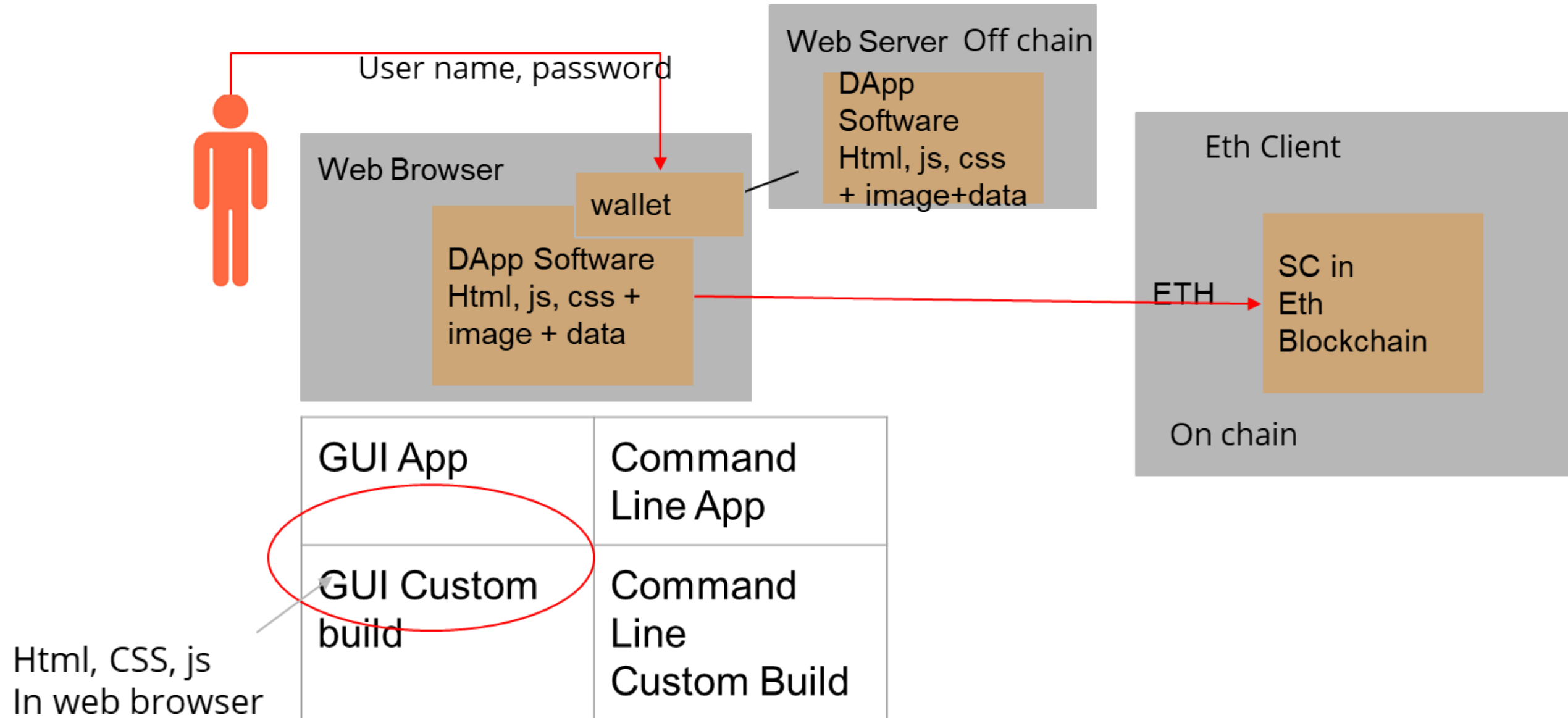
Real Estate Title Insurance

The title registry system uses blockchain to make title issuance instantaneous. A blockchain is a form of digitally distributed ledger that forms a database. Its main characteristic is that it is immutable. Therefore, the information entered in the ledger cannot be deleted, modified, or altered. Blockchain technology has the power to wipe out any third-party intermediary that works for the real estate industry.

The first application used blockchain to tokenize ownership interests in real estate. Digital tokens that prove an investor's ownership of an entity have been offered to investors in real estate funds and joint ventures to finance commercial real estate projects (tokenization). With the tokenization of equity, a blockchain-based ledger system can replace any entity's ownership share registered within the capital stack of a real estate project.

Implementation/ Demonstration

Eth Software Application - Run (Call)



Developing Ethereum Dapps with Truffle, Ganache and MetaMask

Plugin : Metamask

Framework : Truffle

Server environment : NodeJS

Smart Contract Language : Solidity

Library : Web3.js

Ethereum Blockchain Simulator : Ganache



Kenneth Hu

MetaCoin Example Truffle Dapp

You have 2000 META

Send

Amount:

e.g., 95

To Address:

e.g., 0x93e66d9baea28c17d9fc393b53e3fbbd76899dae

Send MetaCoin



MetaMask



Blockchain

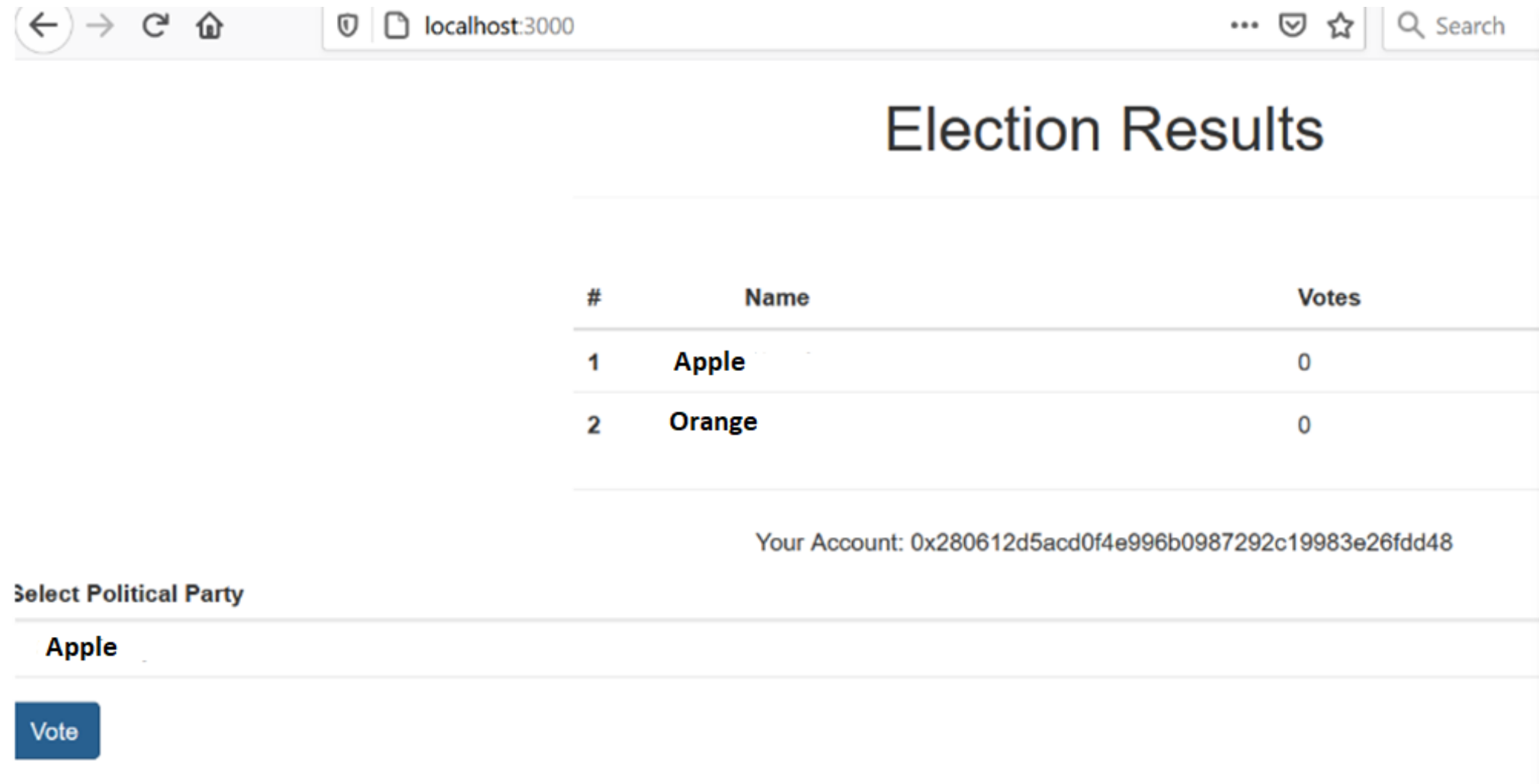
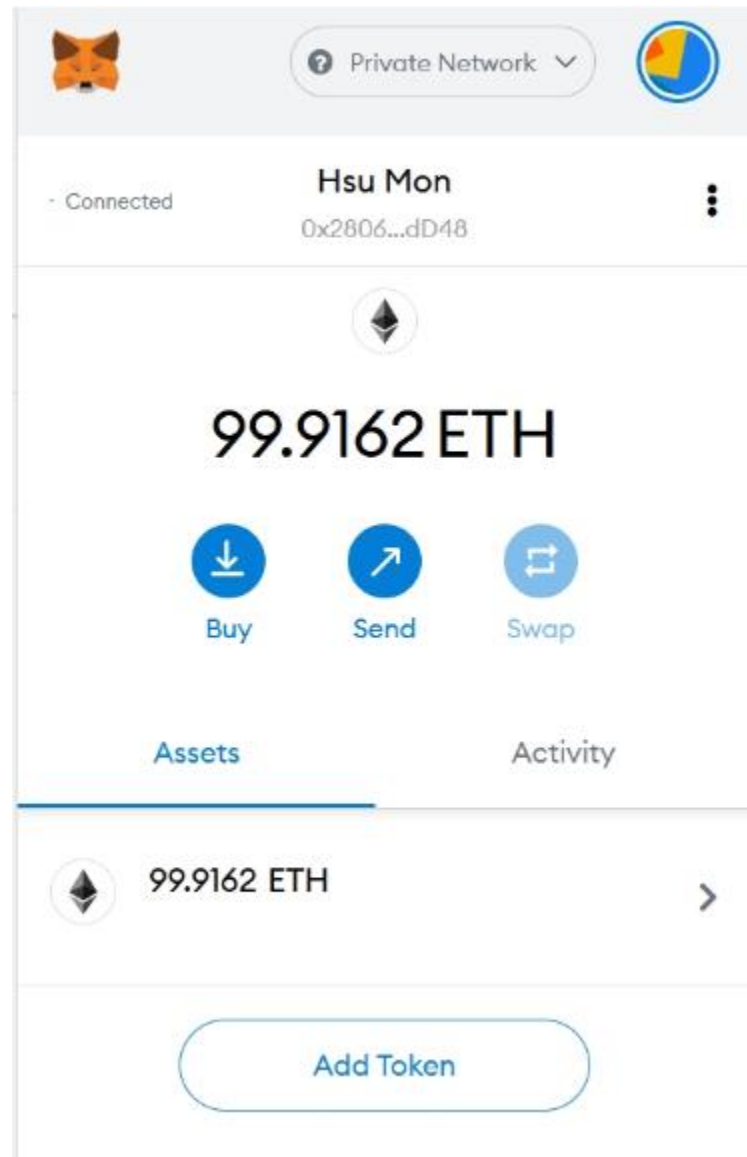


Ganache

Kenneth Hu

- building web apps with the Ethereum blockchain, use the [Truffle web framework](#)
- It [compiles your blockchain contracts](#), injects them into your web app
- With [Metamask](#), all your users need to do is install our Chrome plugin, and they will have their [own secure blockchain accounts right](#) there in the convenience of their browsers

DApps Architecture



DApps Architecture

The screenshot shows a web browser at `localhost:3000` displaying an 'Election Results' page. The page features a table with two candidates, 'Apple' and 'Orange', both with 0 votes. Below the table, the user's account address is shown as `0x280612d5acd0f4e996b0987292c19983e26fdd48`. A 'Select Political Party' section shows 'Apple' selected, with a 'Vote' button below it.

A MetaMask transaction confirmation overlay is visible on the right. It shows the transaction details for a 'VOTE' action on the `http://localhost:3000` network. The transaction is initiated by 'Hsu Mon' and sent to the address `0xe42b...99...`. The gas fee is `0.001988` ETH, and the total amount is also `0.001988` ETH. The overlay includes a 'Reject' button and a 'Confirm' button.

#	Name	Votes
1	Apple	0
2	Orange	0

Your Account: `0x280612d5acd0f4e996b0987292c19983e26fdd48`

Select Political Party

Apple

Vote

MetaMask Notification — Mozilla Fi...

Private Network

Hsu Mon → 0xe42b...99...

`http://localhost:3000`

VOTE

0

DETAILS DATA

GAS FEE `0.001988`

No Conversion Rate Available

Gas Price (GWEI) 20 Gas Limit 99400

AMOUNT + GAS FEE

TOTAL `0.001988`

No Conversion Rate Available

Reject Confirm

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